

Simultaneous Equations (Quadratic)

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Simultaneous Equations — Quadratic (ref ise5)

This document contains all 50 *Simultaneous Equations* questions, each with a fully worked solution and a TikZ graph showing the two equations and their point(s) of intersection — the geometric meaning of the solution. Lines, parabolas, circles (drawn to scale) and hyperbolas are plotted, with tangency, parallel and coincident cases shown where relevant.

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Simultaneous Equations (Quadratic) 01 (ise5-001)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = x + 1$ and $y = x^2 - 1$.

Worked solution.

Step 1: Label the equations and choose a strategy.

$$\textcircled{1} y = x + 1 \quad \textcircled{2} y = x^2 - 1$$

Both equations are already in the form " $y = \text{something}$ ". Setting the two right-hand sides equal removes y and gives a single quadratic in x .

Step 2: Equate the two expressions for y .

$$x + 1 = x^2 - 1$$

Because y equals each right-hand side, those right-hand sides must equal each other at any point of intersection.

Step 3: Rearrange into the standard form $ax^2 + bx + c = 0$.

$$x^2 - x - 2 = 0$$

Move every term to one side so the quadratic equals zero, which is the form needed for factorising.

Step 4: Factorise and solve for x .

$$(x - 2)(x + 1) = 0 \implies x = 2 \text{ or } x = -1$$

Two distinct roots tell you the line cuts the curve in two points, so you expect two coordinate pairs.

Step 5: Substitute each x into the LINEAR equation ① to find y .

$$x = 2 \Rightarrow y = 3; \quad x = -1 \Rightarrow y = 0$$

Always substitute back into the linear equation — it cannot introduce extra (extraneous) solutions, whereas a quadratic substitution can.

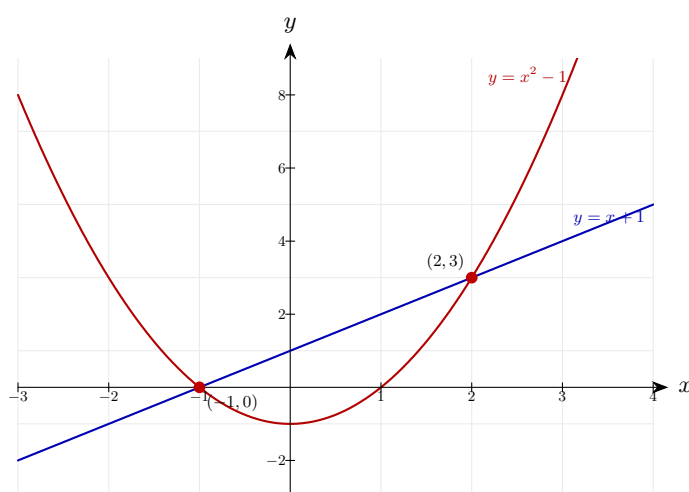
Step 6: State the solution as coordinate pairs.

$$(x, y) = (2, 3) \text{ and } (-1, 0)$$

The final answer must be pairs, not just x -values. Each x is matched with its own y .

Final answer: $(x, y) = (2, 3)$ and $(x, y) = (-1, 0)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 02 (ise5-002)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = 2x + 3$ and $y = x^2$.

Worked solution.

Step 1: Equate the two expressions for y .

$$x^2 = 2x + 3$$

Since both equations are solved for y , the right-hand sides must agree wherever the graphs meet.

Step 2: Bring all terms to one side.

$$x^2 - 2x - 3 = 0$$

You want the quadratic in the standard form $ax^2 + bx + c = 0$ before factorising.

Step 3: Factorise.

$$(x - 3)(x + 1) = 0 \implies x = 3 \text{ or } x = -1$$

Look for two numbers that multiply to -3 and add to -2 : -3 and $+1$.

Step 4: Find each y using the linear equation.

$$x = 3 \Rightarrow y = 2(3) + 3 = 9; \quad x = -1 \Rightarrow y = 2(-1) + 3 = 1$$

Use the linear equation to avoid the risk of generating spurious solutions.

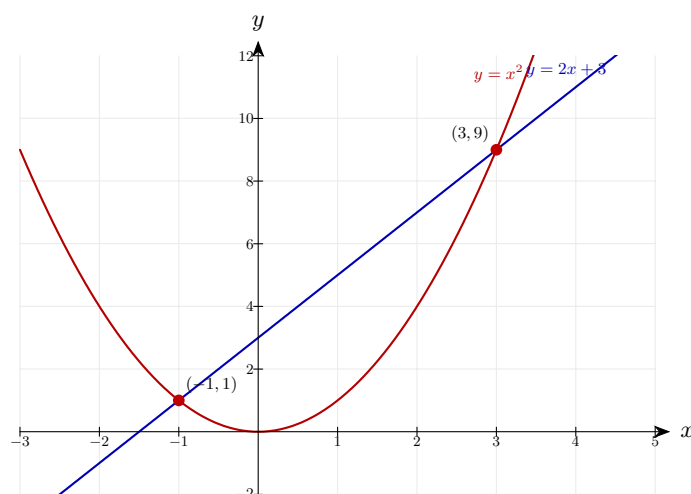
Step 5: Verify in the quadratic equation.

$$3^2 = 9\checkmark; \quad (-1)^2 = 1\checkmark$$

Both pairs satisfy $y = x^2$, so they are genuine intersection points.

Final answer: $(x, y) = (3, 9)$ and $(x, y) = (-1, 1)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 03 (ise5-003)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = x - 3$ and $y = x^2 - 5x + 3$.

Worked solution.

Step 1: Equate the right-hand sides.

$$x - 3 = x^2 - 5x + 3$$

Eliminate y by setting the two expressions for y equal.

Step 2: Rearrange to standard quadratic form.

$$0 = x^2 - 6x + 6$$

Subtract $x - 3$ from both sides: $-5x - x = -6x$ and $3 - (-3) = 6$.

Step 3: Use the quadratic formula because this does not factorise.

$$x = \frac{6 \pm \sqrt{36 - 24}}{2} = \frac{6 \pm \sqrt{12}}{2} = 3 \pm \sqrt{3}$$

Simplify $\sqrt{12} = 2\sqrt{3}$, then divide numerator and denominator by 2.

Step 4: Substitute each x into $y = x - 3$.

$$x = 3 + \sqrt{3} \Rightarrow y = \sqrt{3}; \quad x = 3 - \sqrt{3} \Rightarrow y = -\sqrt{3}$$

The constants cancel cleanly because the line passes through $(3, 0)$.

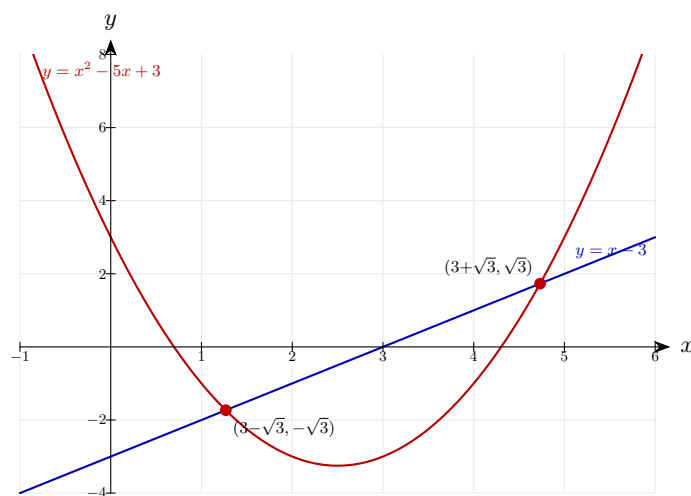
Step 5: State the two intersection points.

$$(3 + \sqrt{3}, \sqrt{3}) \text{ and } (3 - \sqrt{3}, -\sqrt{3})$$

Irrational answers are fine — leave them as exact surds rather than rounding.

Final answer: $(x, y) = (3 + \sqrt{3}, \sqrt{3})$ and $(x, y) = (3 - \sqrt{3}, -\sqrt{3})$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 04 (ise5-004)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = 3x - 2$ and $y = x^2$.

Worked solution.

Step 1: Equate the two expressions for y .

$$x^2 = 3x - 2$$

Both right-hand sides equal y , so they must equal each other at the intersection.

Step 2: Rearrange.

$$x^2 - 3x + 2 = 0$$

Standard form with $a = 1$, $b = -3$, $c = 2$.

Step 3: Factorise.

$$(x - 1)(x - 2) = 0 \implies x = 1 \text{ or } x = 2$$

Two numbers multiplying to 2 and summing to -3 : -1 and -2 .

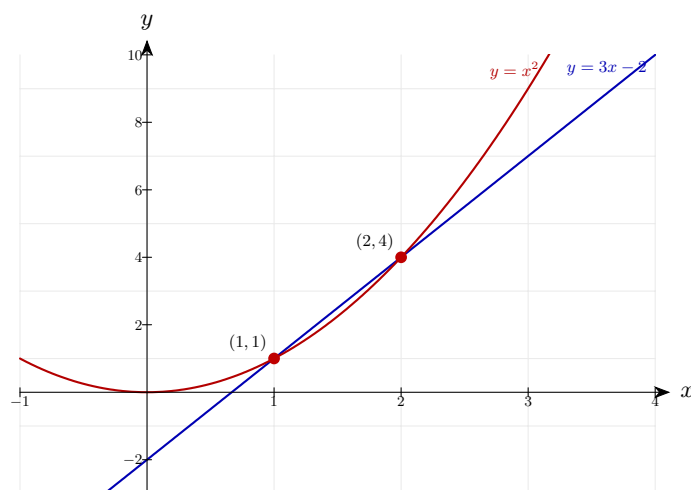
Step 4: Find each corresponding y .

$$x = 1 \Rightarrow y = 1; \quad x = 2 \Rightarrow y = 4$$

Use $y = 3x - 2$: for $x = 1$, $y = 1$; for $x = 2$, $y = 4$. Quickly check: $1^2 = 1$ and $2^2 = 4$ — both consistent with the quadratic.

Final answer: $(x, y) = (1, 1)$ and $(x, y) = (2, 4)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 05 (ise5-005)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = 4x$ and $y = x^2 + 3$.

Worked solution.

Step 1: Equate the two expressions for y .

$$x^2 + 3 = 4x$$

Setting the right-hand sides equal eliminates y and gives a single quadratic in x .

Step 2: Rearrange.

$$x^2 - 4x + 3 = 0$$

Subtract $4x$ from both sides.

Step 3: Factorise.

$$(x - 1)(x - 3) = 0 \implies x = 1 \text{ or } x = 3$$

A standard "find two numbers that multiply to 3, add to -4 " factorisation: $-1, -3$.

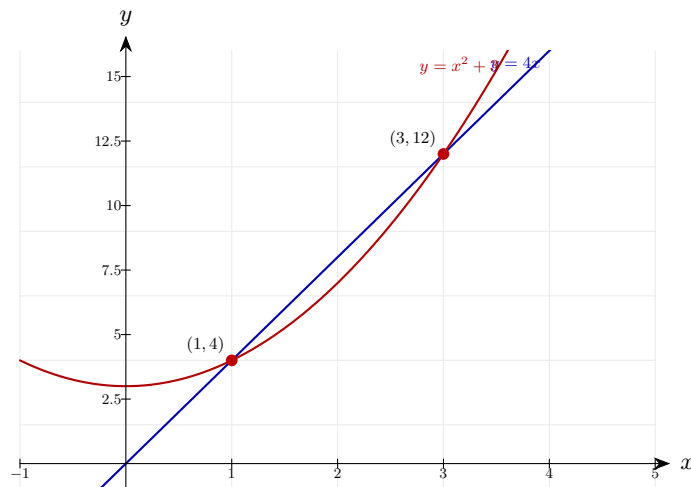
Step 4: Find y from the linear equation.

$$x = 1 \Rightarrow y = 4; \quad x = 3 \Rightarrow y = 12$$

Substitute back into $y = 4x$ — the linear form is simpler and cannot create extra roots.

Final answer: $(x, y) = (1, 4)$ and $(x, y) = (3, 12)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 06 (ise5-006)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = 2x - 1$ and $y = x^2 - 4$.

Worked solution.

Step 1: Equate the two expressions for y .

$$x^2 - 4 = 2x - 1$$

Both sides equal y , so they equal each other at any intersection point.

Step 2: Rearrange to standard form.

$$x^2 - 2x - 3 = 0$$

Move $2x - 1$ to the left: $-4 - (-1) = -3$, $-(2x) = -2x$.

Step 3: Factorise.

$$(x - 3)(x + 1) = 0 \implies x = 3 \text{ or } x = -1$$

Numbers multiplying to -3 and summing to -2 : -3 and $+1$.

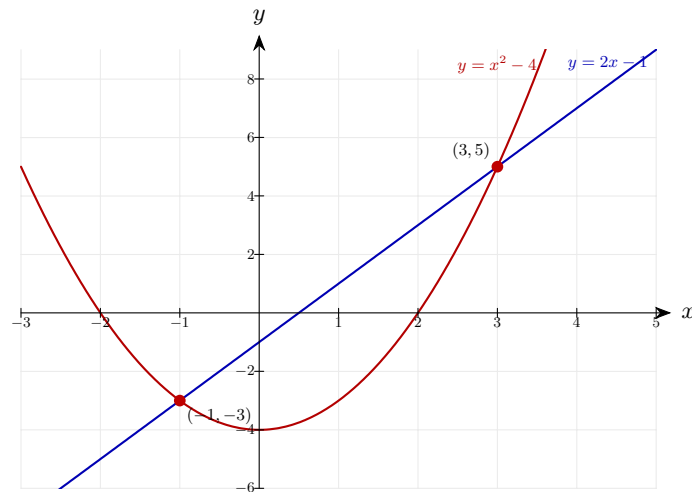
Step 4: Substitute into the linear equation.

$$x = 3 \Rightarrow y = 5; \quad x = -1 \Rightarrow y = -3$$

Pair each x with its own y — a common pitfall is to mix them up.

Final answer: $(x, y) = (3, 5)$ and $(x, y) = (-1, -3)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 07 (ise5-007)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $x + y = 5$ and $y = x^2 - 1$.

Worked solution.

Step 1: Rearrange the linear equation to make y the subject.

$$y = 5 - x$$

Once y is the subject of the linear equation, you can substitute straight into the quadratic.

Step 2: Substitute into the quadratic.

$$5 - x = x^2 - 1$$

Replace y in the second equation with $5 - x$ from the first.

Step 3: Rearrange to standard form.

$$x^2 + x - 6 = 0$$

Move every term to one side: $-1 - 5 = -6$, $-(-x) = +x$.

Step 4: Factorise.

$$(x + 3)(x - 2) = 0 \implies x = -3 \text{ or } x = 2$$

Two numbers multiplying to -6 and summing to 1 : $+3, -2$.

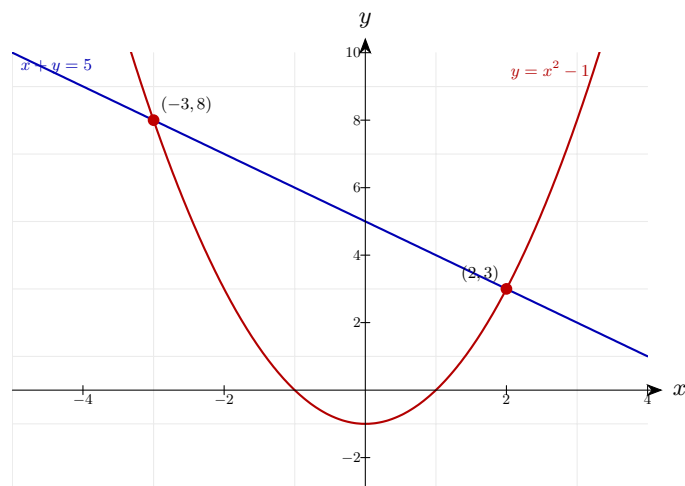
Step 5: Find each y from the linear equation.

$$x = -3 \Rightarrow y = 8; \quad x = 2 \Rightarrow y = 3$$

Always substitute back into $y = 5 - x$ — using the quadratic could introduce extraneous solutions.

Final answer: $(x, y) = (-3, 8)$ and $(x, y) = (2, 3)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 08 (ise5-008)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $2x + y = 7$ and $y = x^2 + 2$.

Worked solution.

Step 1: Make y the subject of the linear equation.

$$y = 7 - 2x$$

Subtract $2x$ from both sides to isolate y .

Step 2: Substitute into the quadratic.

$$7 - 2x = x^2 + 2$$

This produces a single quadratic in x .

Step 3: Rearrange.

$$x^2 + 2x - 5 = 0$$

Bring everything to the left-hand side: $2 - 7 = -5$, $+2x$.

Step 4: Apply the quadratic formula.

$$x = \frac{-2 \pm \sqrt{4 + 20}}{2} = \frac{-2 \pm \sqrt{24}}{2} = -1 \pm \sqrt{6}$$

Simplify $\sqrt{24} = 2\sqrt{6}$ and cancel the factor of 2.

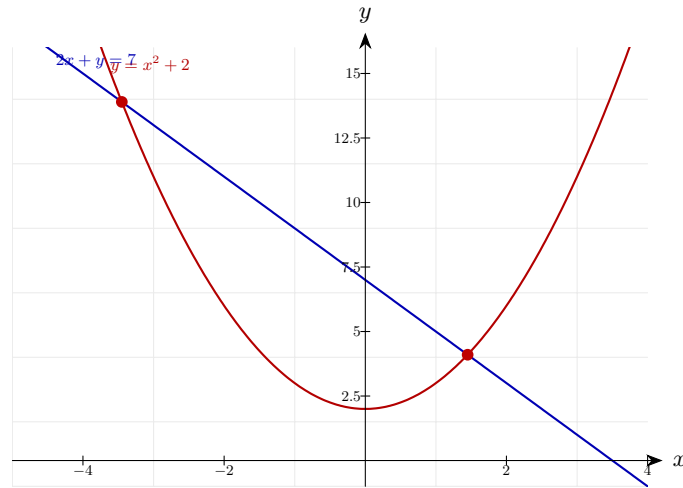
Step 5: Find the corresponding y -values.

$$x = -1 + \sqrt{6} \Rightarrow y = 9 - 2\sqrt{6}; \quad x = -1 - \sqrt{6} \Rightarrow y = 9 + 2\sqrt{6}$$

Use $y = 7 - 2x$: $7 - 2(-1 \pm \sqrt{6}) = 9 \mp 2\sqrt{6}$.

Final answer: $(x, y) = (-1 + \sqrt{6}, 9 - 2\sqrt{6})$ and $(x, y) = (-1 - \sqrt{6}, 9 + 2\sqrt{6})$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 09 (ise5-009)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $x + y = 6$ and $y = x^2$.

Worked solution.

Step 1: Rearrange the linear equation.

$$y = 6 - x$$

Make y the subject so it can be substituted into the quadratic.

Step 2: Substitute.

$$6 - x = x^2$$

Replace y in $y = x^2$ with $6 - x$.

Step 3: Form a quadratic equation.

$$x^2 + x - 6 = 0$$

Move everything to one side; the constant becomes -6 and the x term becomes $+x$.

Step 4: Factorise.

$$(x + 3)(x - 2) = 0 \implies x = -3 \text{ or } x = 2$$

Two numbers with product -6 and sum 1: $+3$ and -2 .

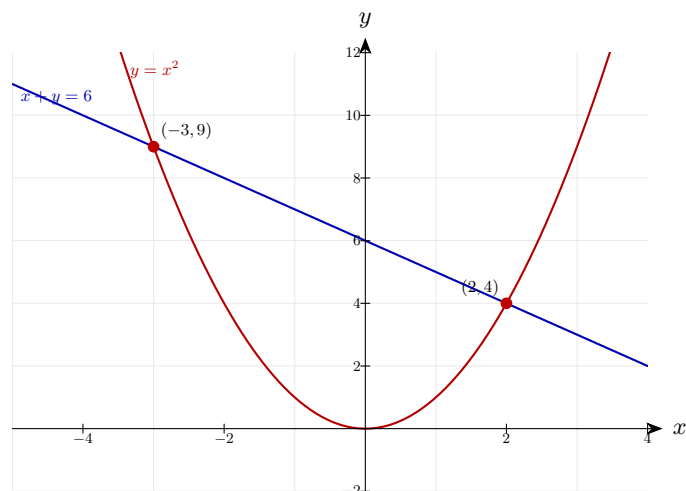
Step 5: Find each y from $y = 6 - x$.

$$x = -3 \Rightarrow y = 9; \quad x = 2 \Rightarrow y = 4$$

Always go back to the linear equation to keep the working safe and the pairing clear.

Final answer: $(x, y) = (-3, 9)$ and $(x, y) = (2, 4)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 10 (ise5-010)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $3x + y = 1$ and $y = 2x^2 - x - 3$.

Worked solution.

Step 1: Make y the subject of the linear equation.

$$y = 1 - 3x$$

Subtract $3x$ from both sides.

Step 2: Substitute into the quadratic.

$$1 - 3x = 2x^2 - x - 3$$

Replace y in the quadratic with the linear expression.

Step 3: Rearrange to standard form.

$$2x^2 + 2x - 4 = 0 \implies x^2 + x - 2 = 0$$

Bring all terms to one side, then divide through by the common factor of 2 to simplify.

Step 4: Factorise.

$$(x + 2)(x - 1) = 0 \implies x = -2 \text{ or } x = 1$$

Numbers multiplying to -2 and summing to 1 are $+2$ and -1 .

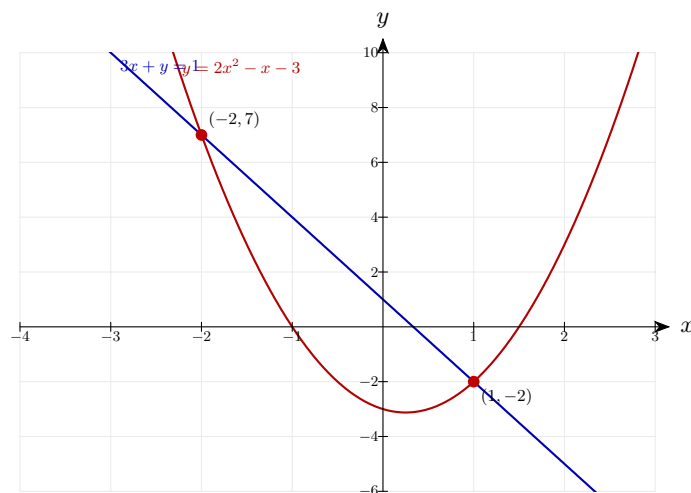
Step 5: Find each y from $y = 1 - 3x$.

$$x = -2 \Rightarrow y = 7; \quad x = 1 \Rightarrow y = -2$$

Use the linear equation to avoid extraneous solutions.

Final answer: $(x, y) = (-2, 7)$ and $(x, y) = (1, -2)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 11 (ise5-011)

Difficulty: Foundation / Marks: 6

Solve the simultaneous equations $y = x + 1$ and $x^2 + y^2 = 25$.

Worked solution.

Step 1: The first equation already has y as the subject.

$$y = x + 1$$

Geometrically, this is a straight line. The second equation is a circle centred at the origin with radius 5.

Step 2: Substitute into the circle.

$$x^2 + (x + 1)^2 = 25$$

Replace y with $x + 1$. Remember to put the bracket round it before squaring.

Step 3: Expand and simplify.

$$x^2 + x^2 + 2x + 1 = 25 \implies 2x^2 + 2x - 24 = 0 \implies x^2 + x - 12 = 0$$

Expand the bracket $(x + 1)^2 = x^2 + 2x + 1$, combine like terms, then divide by 2.

Step 4: Factorise.

$$(x + 4)(x - 3) = 0 \implies x = -4 \text{ or } x = 3$$

Numbers with product -12 and sum 1: $+4$ and -3 .

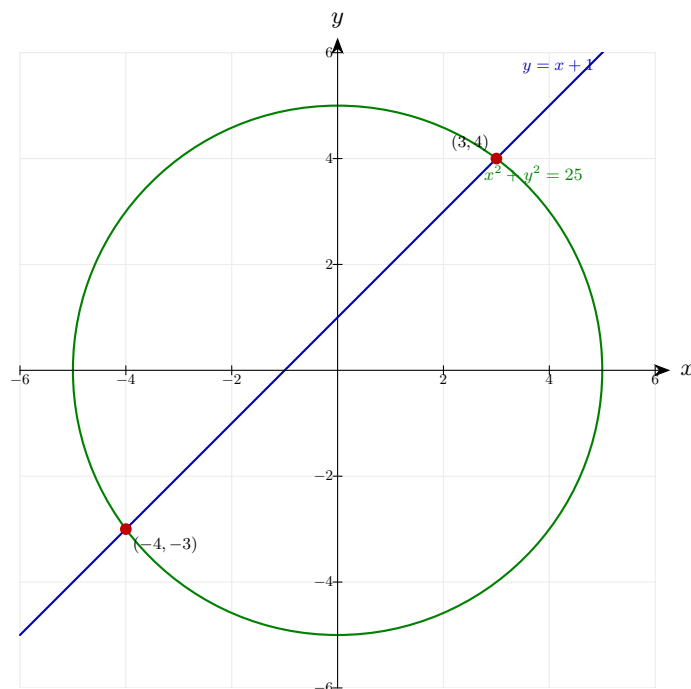
Step 5: Find each y from the linear equation.

$$x = -4 \implies y = -3; \quad x = 3 \implies y = 4$$

Quick sanity check: $(-4)^2 + (-3)^2 = 16 + 9 = 25$ and $3^2 + 4^2 = 9 + 16 = 25$. Both lie on the circle.

Final answer: $(x, y) = (-4, -3)$ and $(x, y) = (3, 4)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 12 (ise5-012)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = 2x$ and $x^2 + y^2 = 20$.

Worked solution.

Step 1: Substitute $y = 2x$ into the circle equation.

$$x^2 + (2x)^2 = 20$$

Square $2x$ to get $4x^2$; make sure the whole $2x$ is squared, not just x .

Step 2: Simplify.

$$5x^2 = 20 \implies x^2 = 4 \implies x = \pm 2$$

A simple quadratic with no linear term. Taking square roots gives two solutions.

Step 3: Find each y using $y = 2x$.

$$x = 2 \Rightarrow y = 4; \quad x = -2 \Rightarrow y = -4$$

Pair each x with its own y . The line through the origin produces symmetric points on the circle.

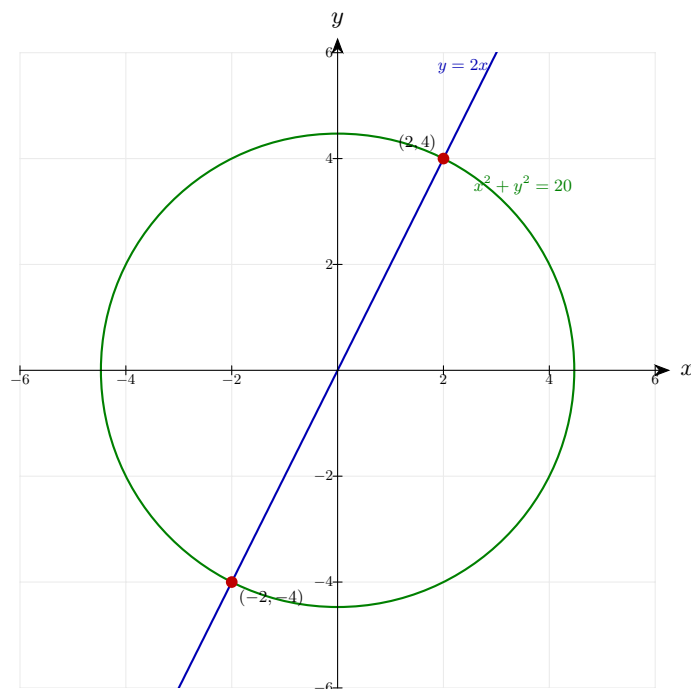
Step 4: Verify.

$$2^2 + 4^2 = 4 + 16 = 20 \checkmark$$

A quick check confirms both points lie on the circle.

Final answer: $(x, y) = (2, 4)$ and $(x, y) = (-2, -4)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 13 (ise5-013)

Difficulty: Foundation / Marks: 6

Solve the simultaneous equations $x + y = 7$ and $x^2 + y^2 = 25$.

Worked solution.

Step 1: Rearrange the linear equation.

$$y = 7 - x$$

Make y the subject so you can substitute it into the circle.

Step 2: Substitute.

$$x^2 + (7 - x)^2 = 25$$

Put the linear expression in brackets and square the whole thing.

Step 3: Expand and simplify.

$$x^2 + 49 - 14x + x^2 = 25 \implies 2x^2 - 14x + 24 = 0 \implies x^2 - 7x + 12 = 0$$

Expand $(7 - x)^2 = 49 - 14x + x^2$, gather terms, then divide by 2.

Step 4: Factorise.

$$(x - 3)(x - 4) = 0 \implies x = 3 \text{ or } x = 4$$

Two numbers with product 12 and sum -7 : -3 and -4 .

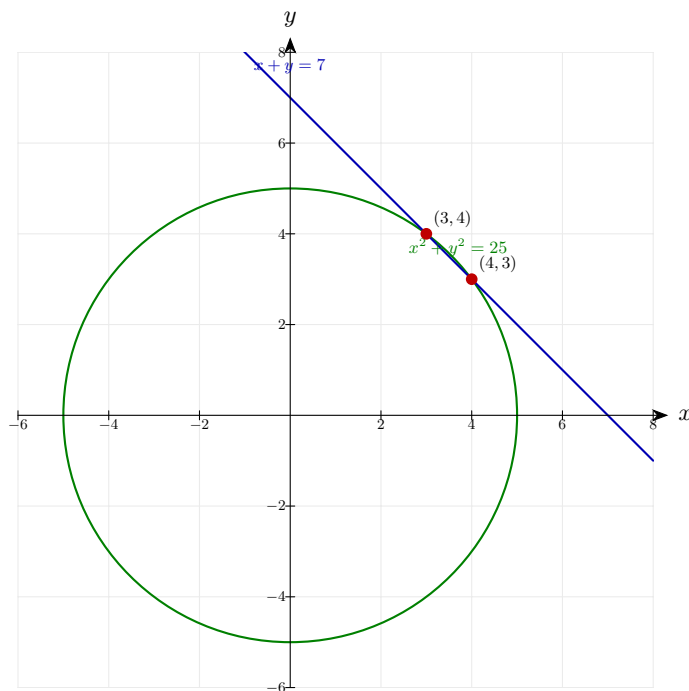
Step 5: Find each y from $y = 7 - x$.

$$x = 3 \implies y = 4; \quad x = 4 \implies y = 3$$

The two points are reflections of each other in the line $y = x$, as expected from the symmetry of the system.

Final answer: $(x, y) = (3, 4)$ and $(x, y) = (4, 3)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 14 (ise5-014)

Difficulty: Foundation / Marks: 6

Solve the simultaneous equations $y = x - 1$ and $x^2 + y^2 = 13$.

Worked solution.

Step 1: Substitute the linear into the circle.

$$x^2 + (x - 1)^2 = 13$$

Bracket the linear expression before squaring to avoid sign errors.

Step 2: Expand and simplify.

$$x^2 + x^2 - 2x + 1 = 13 \implies 2x^2 - 2x - 12 = 0 \implies x^2 - x - 6 = 0$$

$(x - 1)^2 = x^2 - 2x + 1$. After dividing by 2 the numbers become friendlier.

Step 3: Factorise.

$$(x - 3)(x + 2) = 0 \implies x = 3 \text{ or } x = -2$$

Product -6 , sum -1 : -3 and $+2$.

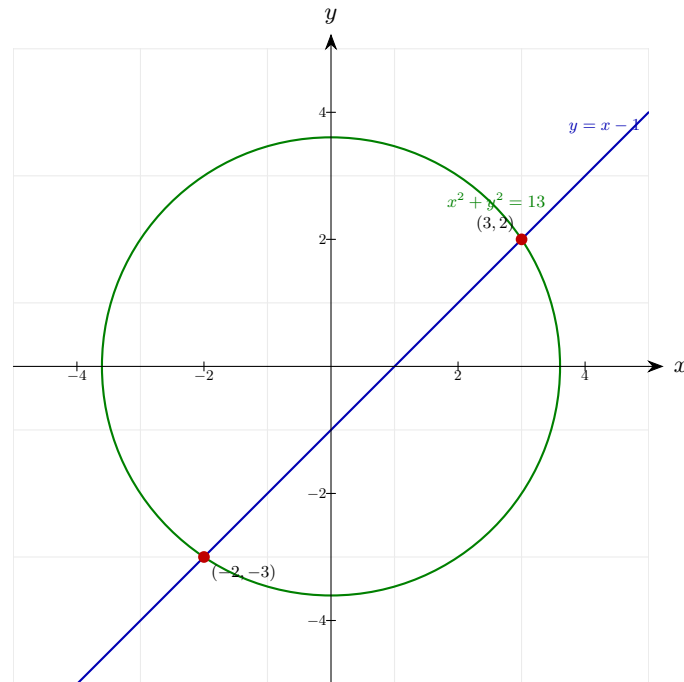
Step 4: Find each y .

$$x = 3 \implies y = 2; \quad x = -2 \implies y = -3$$

Use $y = x - 1$. Check: $3^2 + 2^2 = 13\checkmark$ and $(-2)^2 + (-3)^2 = 13\checkmark$.

Final answer: $(x, y) = (3, 2)$ and $(x, y) = (-2, -3)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 15 (ise5-015)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = x + 2$ and $y = (x - 1)(x + 3)$.

Worked solution.

Step 1: Expand the right-hand side of the quadratic.

$$(x - 1)(x + 3) = x^2 + 2x - 3$$

Expanding first makes the algebra in the next step much cleaner.

Step 2: Equate the two expressions for y .

$$x + 2 = x^2 + 2x - 3$$

Set the right-hand sides equal because both equal y at an intersection point.

Step 3: Rearrange.

$$0 = x^2 + x - 5$$

$2x - x = x$ and $-3 - 2 = -5$.

Step 4: Apply the quadratic formula.

$$x = \frac{-1 \pm \sqrt{1 + 20}}{2} = \frac{-1 \pm \sqrt{21}}{2}$$

Discriminant $1 - 4(1)(-5) = 21$, which is not a perfect square, so the answers are irrational.

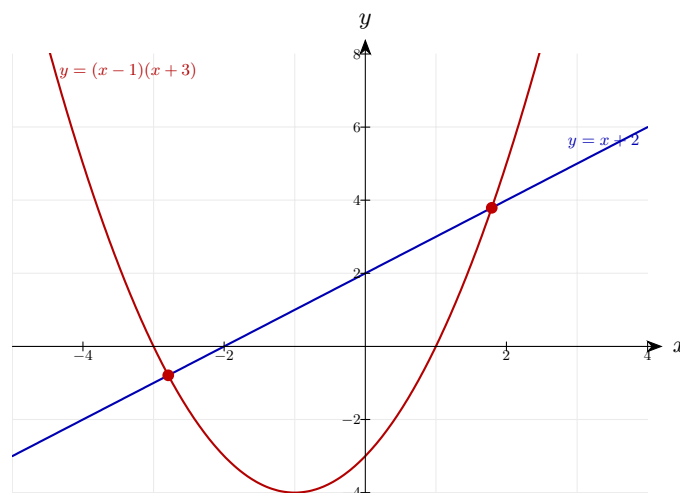
Step 5: Find each corresponding y .

$$y = x + 2 = \frac{3 \pm \sqrt{21}}{2}$$

Add 2 to each x -value: $\frac{-1 \pm \sqrt{21}}{2} + 2 = \frac{3 \pm \sqrt{21}}{2}$.

Final answer: $(x, y) = \left(\frac{-1 + \sqrt{21}}{2}, \frac{3 + \sqrt{21}}{2} \right)$ and $\left(\frac{-1 - \sqrt{21}}{2}, \frac{3 - \sqrt{21}}{2} \right)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 16 (ise5-016)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = 2x + 1$ and $y = (x + 1)(x - 2)$.

Worked solution.

Step 1: Expand the bracketed quadratic.

$$(x + 1)(x - 2) = x^2 - x - 2$$

Use FOIL: $x \cdot x + x \cdot (-2) + 1 \cdot x + 1 \cdot (-2) = x^2 - x - 2$.

Step 2: Set the two expressions for y equal.

$$2x + 1 = x^2 - x - 2$$

Eliminate y by setting the right-hand sides equal.

Step 3: Rearrange.

$$0 = x^2 - 3x - 3$$

$-x - 2x = -3x$ and $-2 - 1 = -3$.

Step 4: Apply the quadratic formula.

$$x = \frac{3 \pm \sqrt{9 + 12}}{2} = \frac{3 \pm \sqrt{21}}{2}$$

Discriminant $9 - 4(1)(-3) = 21$. Surd answers are perfectly acceptable.

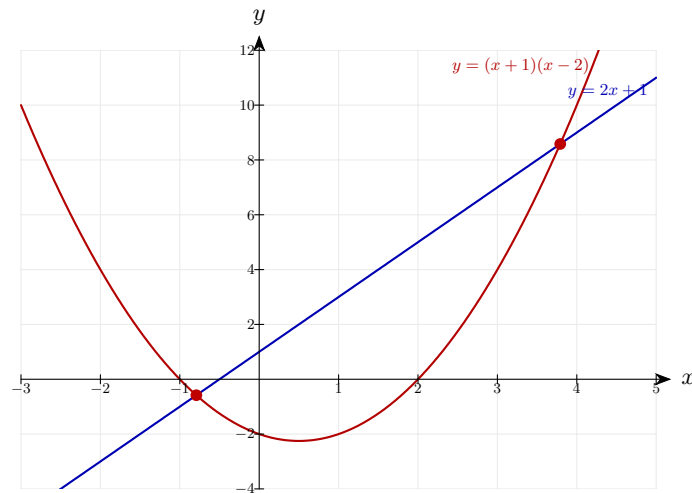
Step 5: Find each corresponding y from $y = 2x + 1$.

$$y = 2 \cdot \frac{3 \pm \sqrt{21}}{2} + 1 = 3 \pm \sqrt{21} + 1 = 4 \pm \sqrt{21}$$

The factor of 2 cancels neatly with the denominator.

Final answer: $(x, y) = \left(\frac{3 + \sqrt{21}}{2}, 4 + \sqrt{21}\right)$ and $\left(\frac{3 - \sqrt{21}}{2}, 4 - \sqrt{21}\right)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 17 (ise5-017)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = 5 - x$ and $y = (x - 1)(x - 2)$.

Worked solution.

Step 1: Expand the quadratic.

$$(x - 1)(x - 2) = x^2 - 3x + 2$$

$$(-1)(-2) = +2 \text{ and } -x - 2x = -3x.$$

Step 2: Equate the two expressions for y .

$$5 - x = x^2 - 3x + 2$$

Both right-hand sides equal y , so they must equal one another.

Step 3: Rearrange.

$$x^2 - 2x - 3 = 0$$

$$-3x - (-x) = -2x, 2 - 5 = -3.$$

Step 4: Factorise.

$$(x - 3)(x + 1) = 0 \implies x = 3 \text{ or } x = -1$$

Product -3 , sum -2 .

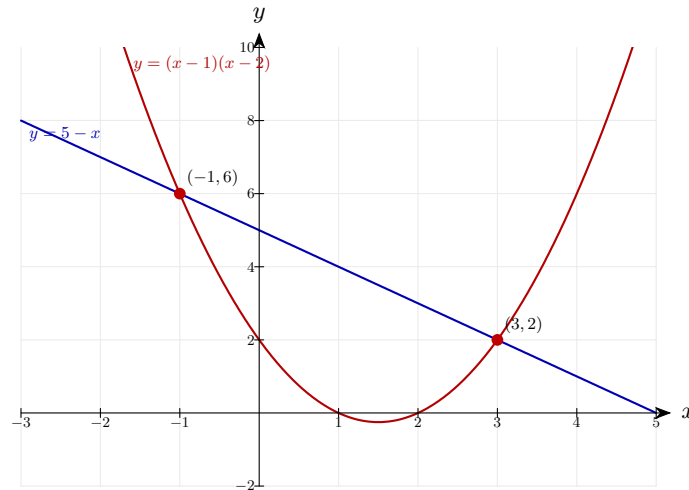
Step 5: Find each y from the linear equation.

$$x = 3 \Rightarrow y = 2; \quad x = -1 \Rightarrow y = 6$$

Use $y = 5 - x$ — the linear equation cannot create false solutions.

Final answer: $(x, y) = (3, 2)$ and $(x, y) = (-1, 6)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 18 (ise5-018)

Difficulty: Foundation / Marks: 5

Solve the simultaneous equations $y = 4 - x$ and $y = (x + 2)(x - 1)$.

Worked solution.

Step 1: Expand the quadratic.

$$(x + 2)(x - 1) = x^2 + x - 2$$

$$2 \cdot (-1) = -2 \text{ and } -x + 2x = x.$$

Step 2: Set the right-hand sides equal.

$$4 - x = x^2 + x - 2$$

Equate both expressions for y .

Step 3: Rearrange.

$$x^2 + 2x - 6 = 0$$

$$x - (-x) = 2x, \quad -2 - 4 = -6.$$

Step 4: Apply the quadratic formula.

$$x = \frac{-2 \pm \sqrt{4 + 24}}{2} = \frac{-2 \pm \sqrt{28}}{2} = -1 \pm \sqrt{7}$$

Simplify $\sqrt{28} = 2\sqrt{7}$ and cancel the 2.

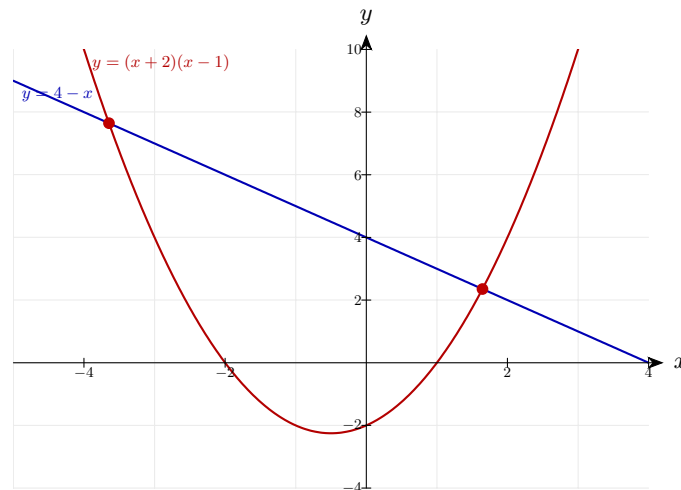
Step 5: Find each y from $y = 4 - x$.

$$y = 4 - (-1 \pm \sqrt{7}) = 5 \mp \sqrt{7}$$

The sign of the surd flips when subtracting.

Final answer: $(x, y) = (-1 + \sqrt{7}, 5 - \sqrt{7})$ and $(-1 - \sqrt{7}, 5 + \sqrt{7})$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 19 (ise5-019)

Difficulty: Standard / Marks: 6

Solve the simultaneous equations $y = x + 4$ and $y = x^2 + 2x - 1$, giving your answers as exact surds.

Worked solution.

Step 1: Equate the two expressions for y .

$$x + 4 = x^2 + 2x - 1$$

Set the linear equal to the quadratic at any intersection point.

Step 2: Rearrange.

$$x^2 + x - 5 = 0$$

$2x - x = x$ and $-1 - 4 = -5$.

Step 3: Apply the quadratic formula.

$$x = \frac{-1 \pm \sqrt{1 + 20}}{2} = \frac{-1 \pm \sqrt{21}}{2}$$

Discriminant 21 is not a perfect square, so the roots are irrational. Leave them as surds rather than rounding.

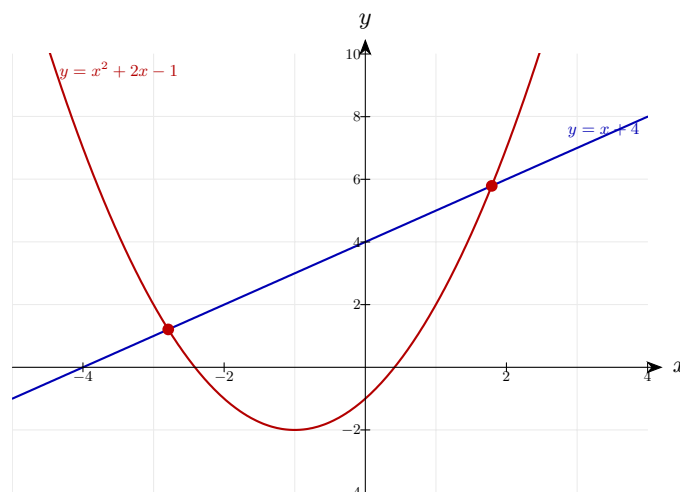
Step 4: Find each y from $y = x + 4$.

$$y = \frac{-1 \pm \sqrt{21}}{2} + 4 = \frac{7 \pm \sqrt{21}}{2}$$

$$4 = \frac{8}{2}, \text{ so adding gives } \frac{-1 + 8 \pm \sqrt{21}}{2} = \frac{7 \pm \sqrt{21}}{2}.$$

Final answer: $(x, y) = \left(\frac{-1 + \sqrt{21}}{2}, \frac{7 + \sqrt{21}}{2} \right)$ and $\left(\frac{-1 - \sqrt{21}}{2}, \frac{7 - \sqrt{21}}{2} \right)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 20 (ise5-020)

Difficulty: Standard / Marks: 6

Solve the simultaneous equations $y = x - 2$ and $y = x^2 - 5$, giving answers as exact surds.

Worked solution.

Step 1: Equate the right-hand sides.

$$x - 2 = x^2 - 5$$

Set the two expressions for y equal at the points of intersection.

Step 2: Rearrange.

$$x^2 - x - 3 = 0$$

Bring all terms to one side: $-5 + 2 = -3$, $-x$ appears with a sign change.

Step 3: Quadratic formula.

$$x = \frac{1 \pm \sqrt{1 + 12}}{2} = \frac{1 \pm \sqrt{13}}{2}$$

Discriminant $1 + 12 = 13$. Not a perfect square, so leave as a surd.

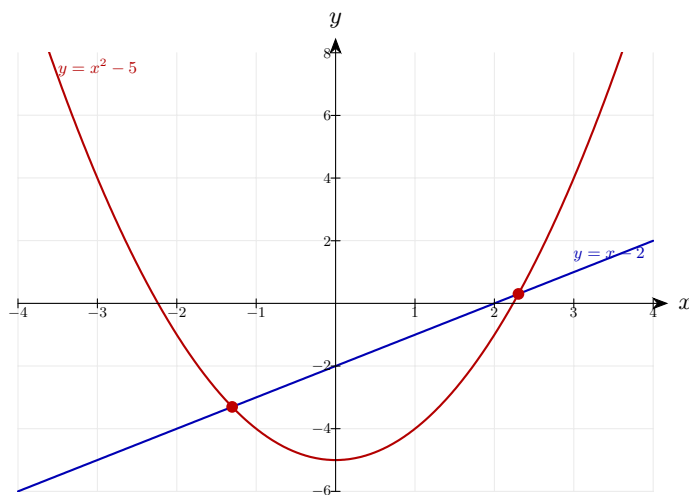
Step 4: Find each y .

$$y = \frac{1 \pm \sqrt{13}}{2} - 2 = \frac{-3 \pm \sqrt{13}}{2}$$

$$-2 = -\frac{4}{2}, \text{ so } \frac{1 - 4 \pm \sqrt{13}}{2}.$$

Final answer: $(x, y) = \left(\frac{1 + \sqrt{13}}{2}, \frac{-3 + \sqrt{13}}{2}\right)$ and $\left(\frac{1 - \sqrt{13}}{2}, \frac{-3 - \sqrt{13}}{2}\right)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 21 (ise5-021)

Difficulty: Standard / Marks: 5

Solve the simultaneous equations $y = 2x$ and $y = x^2 - 3$, giving answers as exact surds.

Worked solution.

Step 1: Equate the expressions for y .

$$2x = x^2 - 3$$

Substituting eliminates y and gives a quadratic in x .

Step 2: Rearrange.

$$x^2 - 2x - 3 = 0$$

Subtract $2x$ from both sides.

Step 3: This actually factorises.

$$(x - 3)(x + 1) = 0 \implies x = 3 \text{ or } x = -1$$

Product -3 , sum -2 . Sometimes the "surd" version actually gives rational answers — always check whether the discriminant is a perfect square first.

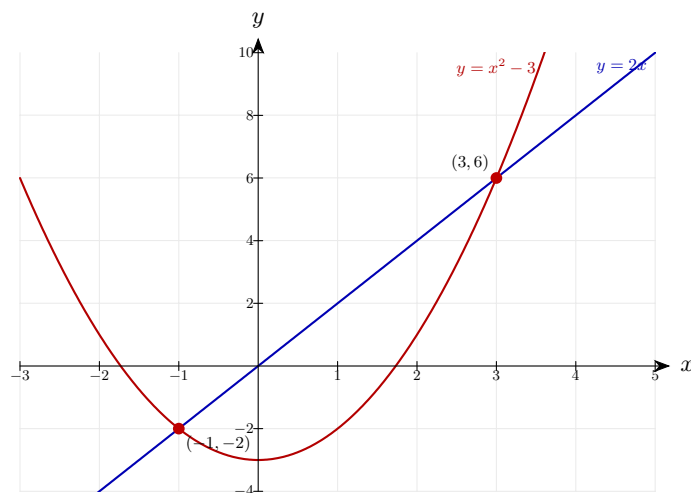
Step 4: Find each y from $y = 2x$.

$$x = 3 \implies y = 6; \quad x = -1 \implies y = -2$$

In this case the answers are integers, so no surds are needed in the final answer.

Final answer: $(x, y) = (3, 6)$ and $(x, y) = (-1, -2)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 22 (ise5-022)

Difficulty: Standard / Marks: 6

Solve the simultaneous equations $y = 3 - 2x$ and $y = x^2 + x - 4$, giving your answers as exact surds.

Worked solution.

Step 1: Equate the right-hand sides.

$$3 - 2x = x^2 + x - 4$$

Both expressions equal y , so they equal one another at any intersection.

Step 2: Rearrange.

$$x^2 + 3x - 7 = 0$$

$$x - (-2x) = 3x \text{ and } -4 - 3 = -7.$$

Step 3: Quadratic formula.

$$x = \frac{-3 \pm \sqrt{9 + 28}}{2} = \frac{-3 \pm \sqrt{37}}{2}$$

Discriminant $9 + 28 = 37$, not a perfect square, so the roots are irrational.

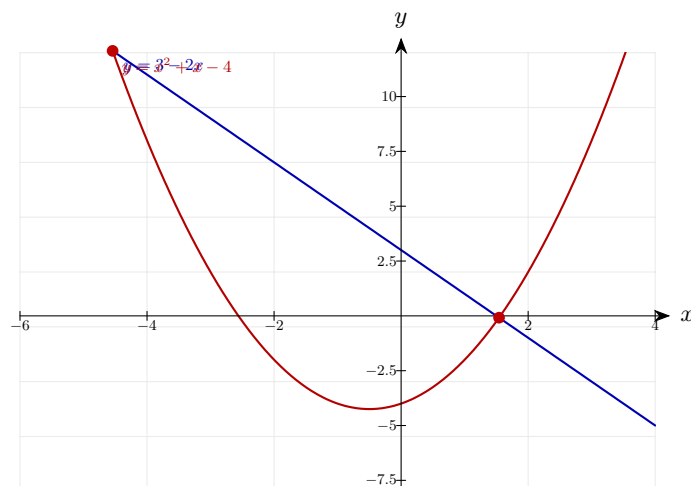
Step 4: Find each y .

$$y = 3 - 2 \cdot \frac{-3 \pm \sqrt{37}}{2} = 3 - (-3 \pm \sqrt{37}) = 6 \mp \sqrt{37}$$

The 2 in front cancels the denominator, then a sign flip occurs because of the minus.

Final answer: $(x, y) = \left(\frac{-3 + \sqrt{37}}{2}, 6 - \sqrt{37} \right)$ and $\left(\frac{-3 - \sqrt{37}}{2}, 6 + \sqrt{37} \right)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 23 (ise5-023)

Difficulty: Standard / Marks: 6

Solve the simultaneous equations $y = 2x - 1$ and $y = x^2$. Interpret your answer geometrically.

Worked solution.

Step 1: Equate the expressions for y .

$$x^2 = 2x - 1$$

Setting the two right-hand sides equal eliminates y .

Step 2: Rearrange.

$$x^2 - 2x + 1 = 0$$

Move every term to the left.

Step 3: Compute the discriminant.

$$b^2 - 4ac = (-2)^2 - 4(1)(1) = 4 - 4 = 0$$

A discriminant of zero means the quadratic has exactly one (repeated) real root, so the line meets the curve at exactly one point.

Step 4: Solve.

$$(x - 1)^2 = 0 \implies x = 1$$

The factorisation is a perfect square.

Step 5: Find y .

$$y = 2(1) - 1 = 1$$

Substitute $x = 1$ into the linear equation.

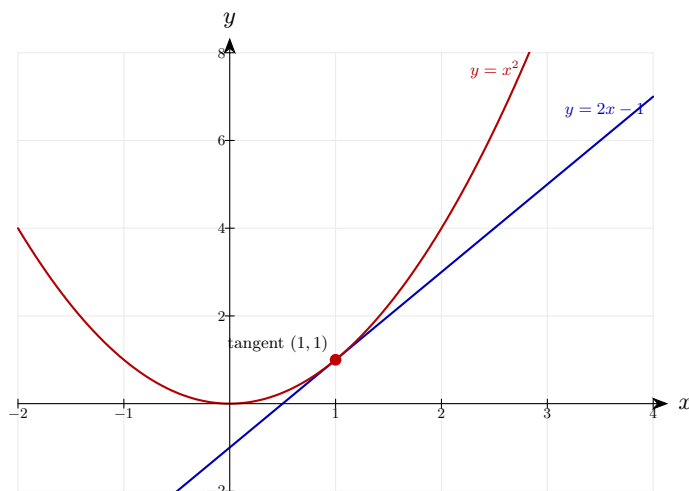
Step 6: Geometric interpretation.

$$\text{Discriminant} = 0 \implies \text{line is tangent to the curve at } (1, 1).$$

Discriminant zero is the algebraic signature of tangency: the line touches the curve at one point without crossing it.

Final answer: $(x, y) = (1, 1)$; the line is tangent to the parabola at this point.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 24 (ise5-024)

Difficulty: Standard / Marks: 6

Show that the line $y = 4x - 4$ is a tangent to the curve $y = x^2$ and find the point of tangency.

Worked solution.

Step 1: Equate the right-hand sides.

$$x^2 = 4x - 4$$

A tangent meets the curve at exactly one (repeated) point — we will check this via the discriminant.

Step 2: Rearrange to standard form.

$$x^2 - 4x + 4 = 0$$

Move everything to one side.

Step 3: Compute the discriminant.

$$b^2 - 4ac = 16 - 16 = 0$$

The discriminant is zero — confirming the line is tangent to the curve.

Step 4: Solve for x .

$$(x - 2)^2 = 0 \implies x = 2$$

A perfect-square factorisation.

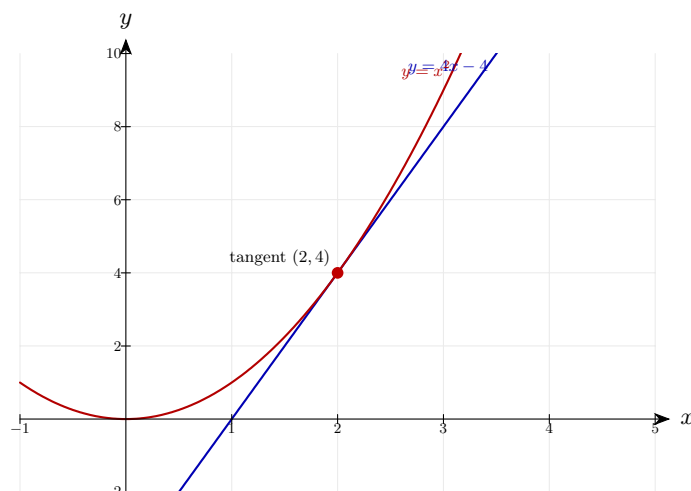
Step 5: Find y .

$$y = 4(2) - 4 = 4$$

Substitute back into the linear equation.

Final answer: Tangent at $(x, y) = (2, 4)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 25 (ise5-025)

Difficulty: Standard / Marks: 6

Show that the line $y = -2x + 1$ is tangent to the curve $y = -x^2$, and find the point of contact.

Worked solution.

Step 1: Set the right-hand sides equal.

$$-x^2 = -2x + 1$$

Both equal y at any point of contact, so the right-hand sides must agree there.

Step 2: Rearrange to standard form.

$$x^2 - 2x + 1 = 0$$

Multiply both sides by -1 (which reverses every sign) and bring all terms to the left so the leading coefficient is positive.

Step 3: Compute the discriminant.

$$b^2 - 4ac = (-2)^2 - 4(1)(1) = 4 - 4 = 0$$

A discriminant of zero is the algebraic signature of tangency — exactly one (repeated) real root, so the line touches the curve at one point.

Step 4: Solve the (perfect-square) quadratic.

$$(x - 1)^2 = 0 \implies x = 1$$

The factorisation is a perfect square, giving the repeated root $x = 1$.

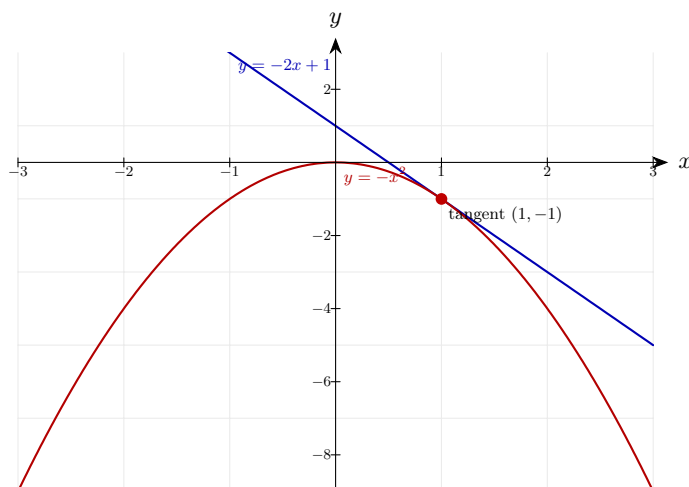
Step 5: Find y .

$$y = -2(1) + 1 = -1$$

Substitute $x = 1$ into the linear equation. Check on the curve: $-1^2 = -1$ ✓.

Final answer: The line is tangent to the curve at $(x, y) = (1, -1)$.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 26 (ise5-026)

Difficulty: Standard / Marks: 6

Show that the line $y = 6x - 9$ is tangent to the curve $y = x^2$, and find the point of tangency.

Worked solution.

Step 1: Equate the two expressions for y .

$$x^2 = 6x - 9$$

Set the linear right-hand side equal to the quadratic right-hand side.

Step 2: Rearrange.

$$x^2 - 6x + 9 = 0$$

A standard form quadratic.

Step 3: Compute the discriminant.

$$b^2 - 4ac = 36 - 36 = 0$$

Discriminant zero confirms tangency — exactly one (repeated) root.

Step 4: Solve.

$$(x - 3)^2 = 0 \implies x = 3$$

A perfect-square trinomial.

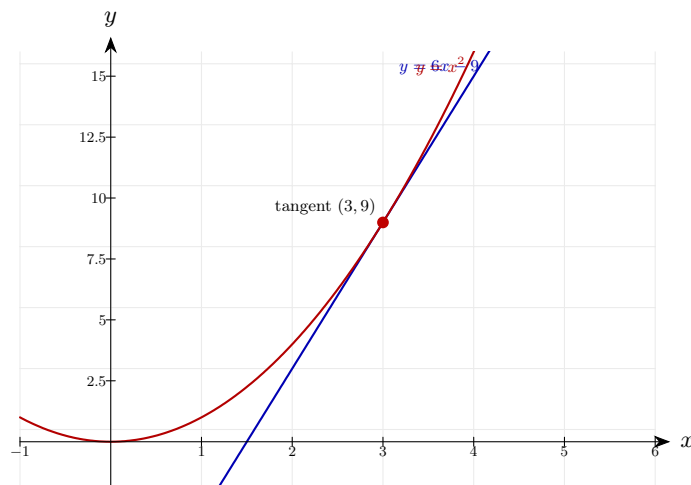
Step 5: Find y .

$$y = 6(3) - 9 = 9$$

Substitute $x = 3$ into the linear equation.

Final answer: The line is tangent at $(x, y) = (3, 9)$.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 27 (ise5-027)

Difficulty: Standard / Marks: 5

Show that the line $y = x - 3$ does not meet the curve $y = x^2 + 1$.

Worked solution.

Step 1: Equate the two expressions for y .

$$x - 3 = x^2 + 1$$

Eliminate y by setting the right-hand sides equal.

Step 2: Rearrange.

$$x^2 - x + 4 = 0$$

$1 + 3 = 4$, and moving x across gives $-x$.

Step 3: Compute the discriminant.

$$b^2 - 4ac = 1 - 16 = -15 < 0$$

A negative discriminant means there are no real solutions, so the line and the curve do not intersect.

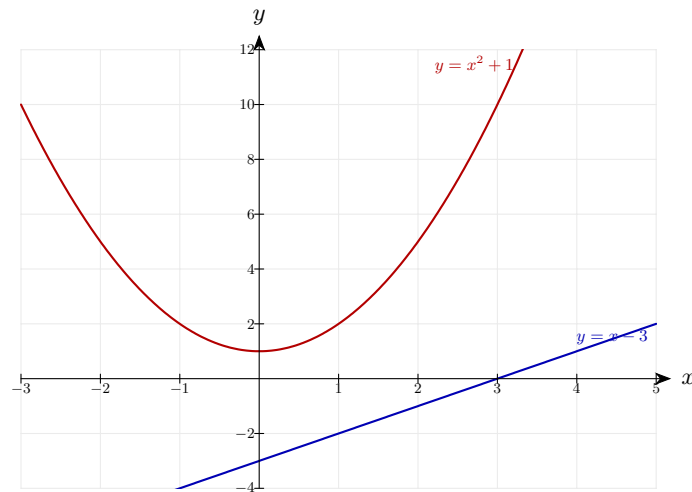
Step 4: Geometric interpretation.

Line misses the curve entirely.

The line passes "under" the parabola; the parabola has minimum value 1 (at $x = 0$), but the line at $x = 0$ gives $y = -3$, and the line never rises fast enough to catch the parabola.

Final answer: No real solutions — discriminant $= -15 < 0$, so the line does not meet the curve.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 28 (ise5-028)

Difficulty: Standard / Marks: 5

Show that the line $y = 2x + 5$ does not intersect the circle $x^2 + y^2 = 1$.

Worked solution.

Step 1: Substitute the line into the circle equation.

$$x^2 + (2x + 5)^2 = 1$$

Replace y in the circle equation with the linear expression so the system reduces to a single quadratic in x .

Step 2: Expand and simplify.

$$x^2 + 4x^2 + 20x + 25 = 1 \implies 5x^2 + 20x + 24 = 0$$

Expand $(2x + 5)^2 = 4x^2 + 20x + 25$, then bring the 1 across to the left-hand side.

Step 3: Compute the discriminant.

$$b^2 - 4ac = 20^2 - 4(5)(24) = 400 - 480 = -80$$

A negative discriminant means the quadratic has no real solutions, so there is no real x at which the line meets the circle.

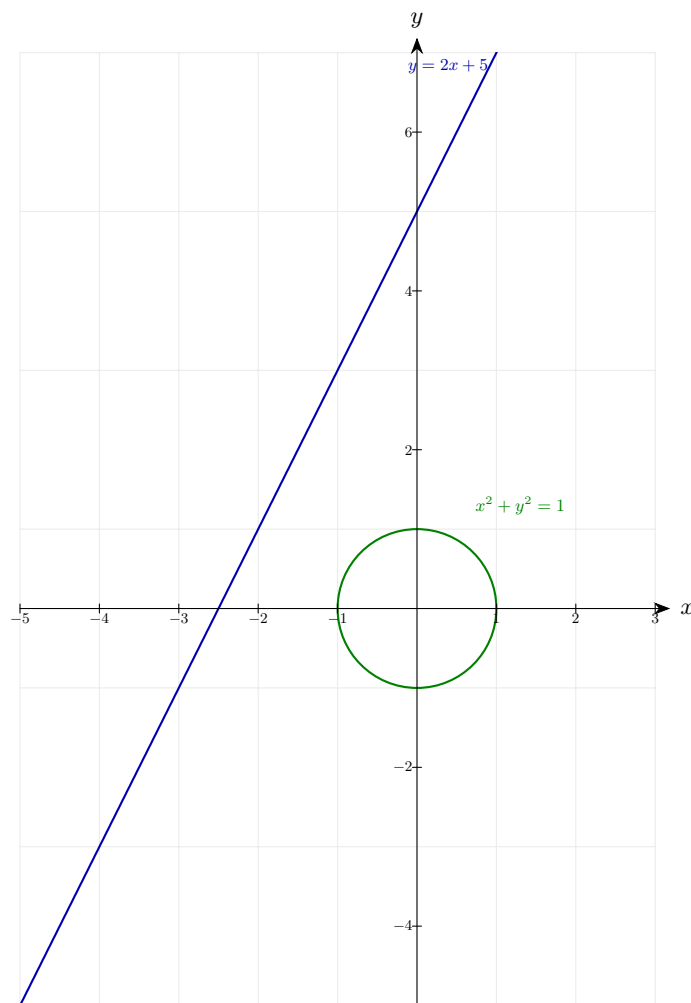
Step 4: Geometric interpretation.

$$\text{discriminant} < 0 \implies \text{no intersection.}$$

The perpendicular distance from the origin to the line $2x - y + 5 = 0$ is $\frac{5}{\sqrt{5}} = \sqrt{5} > 1$, which exceeds the radius — confirming the line lies entirely outside the unit circle.

Final answer: No real intersection — discriminant $= -80 < 0$, so the line does not meet the circle.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 29 (ise5-029)

Difficulty: Standard / Marks: 5

Show that the line $y = -x - 5$ does not meet the curve $y = x^2 - 3x + 2$.

Worked solution.

Step 1: Equate the expressions for y .

$$-x - 5 = x^2 - 3x + 2$$

Set the two right-hand sides equal.

Step 2: Rearrange.

$$x^2 - 2x + 7 = 0$$

$$-3x - (-x) = -2x, 2 - (-5) = 7.$$

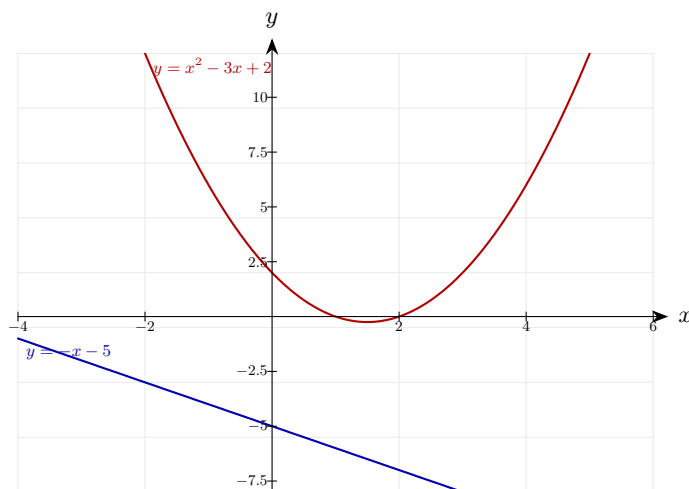
Step 3: Compute the discriminant.

$$b^2 - 4ac = 4 - 28 = -24 < 0$$

Negative discriminant — there are no real roots, so the line and curve do not meet.

Final answer: No real intersection; discriminant $= -24 < 0$.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 30 (ise5-030)

Difficulty: Standard / Marks: 5

Show that the line $y = 3x + 4$ does not intersect the curve $y = x^2 + x + 6$.

Worked solution.

Step 1: Equate the right-hand sides.

$$3x + 4 = x^2 + x + 6$$

Eliminate y by setting the two expressions for y equal at any potential intersection.

Step 2: Rearrange to standard form.

$$x^2 - 2x + 2 = 0$$

Move everything to one side: $x - 3x = -2x$ and $6 - 4 = 2$.

Step 3: Compute the discriminant.

$$b^2 - 4ac = (-2)^2 - 4(1)(2) = 4 - 8 = -4$$

A negative discriminant means the quadratic has no real roots, so there is no real x at which the line meets the curve.

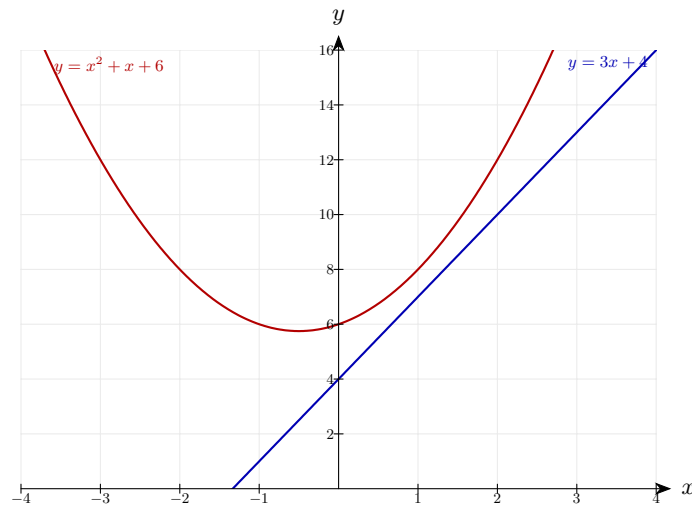
Step 4: Geometric interpretation.

$$\text{discriminant} < 0 \implies \text{line misses the curve entirely.}$$

Completing the square, $x^2 + x + 6 - (3x + 4) = (x - 1)^2 + 1 \geq 1 > 0$ for every real x , so the curve sits strictly above the line.

Final answer: No real intersection — discriminant $= -4 < 0$, so the line does not meet the curve.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 31 (ise5-031)

Difficulty: Standard / Marks: 5

Solve the simultaneous equations $x + y = 5$ and $xy = 6$.

Worked solution.

Step 1: Rearrange the linear equation.

$$y = 5 - x$$

Make y the subject of the simpler equation.

Step 2: Substitute into $xy = 6$.

$$x(5 - x) = 6 \implies 5x - x^2 = 6$$

Distribute, then bring everything to one side.

Step 3: Rearrange and factorise.

$$x^2 - 5x + 6 = 0 \implies (x - 2)(x - 3) = 0 \implies x = 2 \text{ or } x = 3$$

Two numbers with product 6 and sum -5 : $-2, -3$.

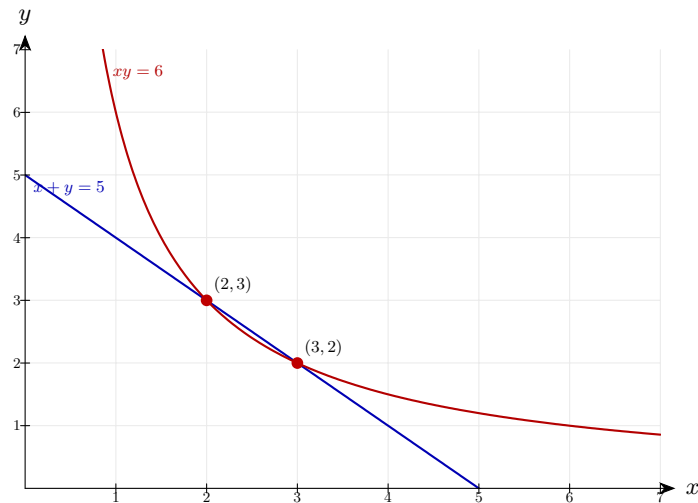
Step 4: Find each y from $y = 5 - x$.

$$x = 2 \implies y = 3; \quad x = 3 \implies y = 2$$

The system is symmetric in x and y , so swapping their roles gives the other solution.

Final answer: $(x, y) = (2, 3)$ and $(x, y) = (3, 2)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 32 (ise5-032)

Difficulty: Standard / Marks: 5

Solve the simultaneous equations $2x + y = 7$ and $xy = 6$.

Worked solution.

Step 1: Rearrange the linear equation.

$$y = 7 - 2x$$

Make y the subject.

Step 2: Substitute into $xy = 6$.

$$x(7 - 2x) = 6 \implies 7x - 2x^2 = 6$$

Distribute over the bracket.

Step 3: Rearrange and factorise.

$$2x^2 - 7x + 6 = 0 \implies (2x - 3)(x - 2) = 0 \implies x = \frac{3}{2} \text{ or } x = 2$$

Multiply: $2x \cdot x = 2x^2$, $2x \cdot (-2) + (-3) \cdot x = -4x - 3x = -7x$, $(-3)(-2) = 6$. Check.

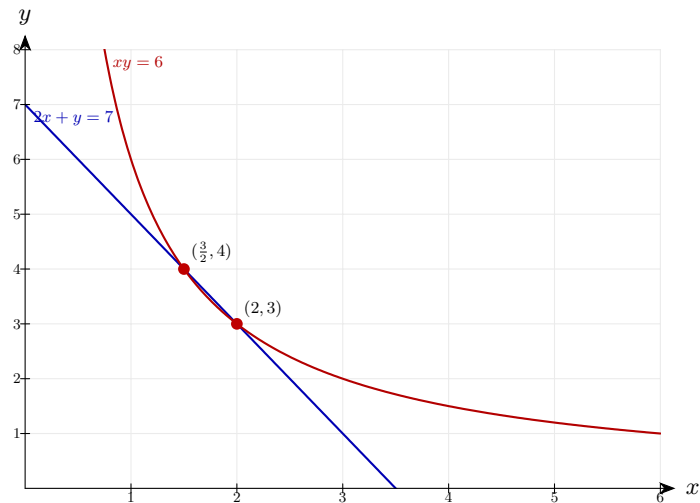
Step 4: Find each y from $y = 7 - 2x$.

$$x = \frac{3}{2} \Rightarrow y = 4; \quad x = 2 \Rightarrow y = 3$$

Quick check: $\frac{3}{2} \times 4 = 6\checkmark$ and $2 \times 3 = 6\checkmark$.

Final answer: $(x, y) = \left(\frac{3}{2}, 4\right)$ and $(x, y) = (2, 3)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 33 (ise5-033)

Difficulty: Standard / Marks: 5

Solve the simultaneous equations $x - y = 1$ and $xy = 12$.

Worked solution.

Step 1: Rearrange.

$$x = y + 1$$

Making x the subject is just as valid as making y the subject.

Step 2: Substitute into $xy = 12$.

$$(y + 1)y = 12 \implies y^2 + y - 12 = 0$$

Expand and bring everything to one side.

Step 3: Factorise.

$$(y + 4)(y - 3) = 0 \implies y = -4 \text{ or } y = 3$$

Two numbers multiplying to -12 and summing to 1 : $+4$ and -3 .

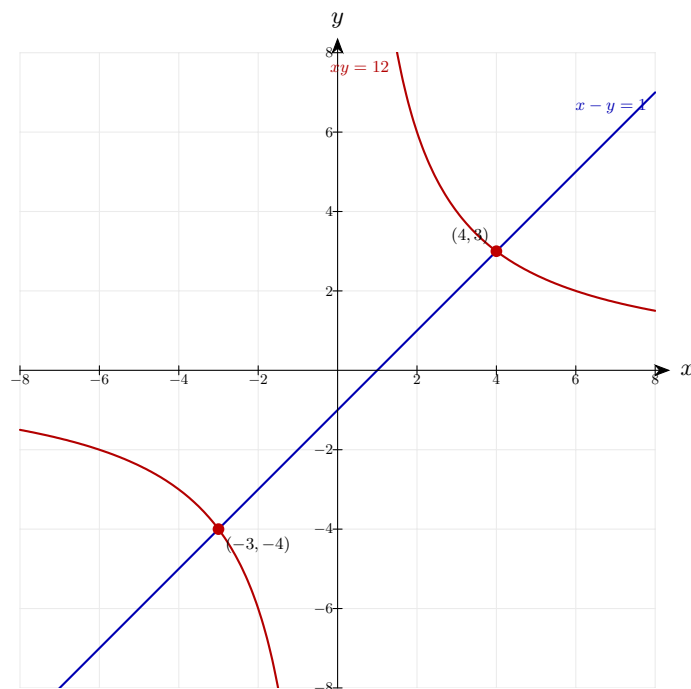
Step 4: Find each x from $x = y + 1$.

$$y = -4 \implies x = -3; \quad y = 3 \implies x = 4$$

Sanity check the product: $(-3)(-4) = 12\checkmark$ and $4 \times 3 = 12\checkmark$.

Final answer: $(x, y) = (-3, -4)$ and $(x, y) = (4, 3)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 34 (ise5-034)

Difficulty: Standard / Marks: 6

Solve the simultaneous equations $3x + 2y = 12$ and $xy = 6$.

Worked solution.

Step 1: Rearrange the linear equation.

$$y = \frac{12 - 3x}{2}$$

Make y the subject.

Step 2: Substitute into $xy = 6$.

$$x \cdot \frac{12 - 3x}{2} = 6 \implies x(12 - 3x) = 12$$

Multiply both sides by 2 to clear the fraction.

Step 3: Expand and rearrange.

$$12x - 3x^2 = 12 \implies 3x^2 - 12x + 12 = 0 \implies x^2 - 4x + 4 = 0$$

Divide through by 3 for cleaner numbers.

Step 4: Solve.

$$(x - 2)^2 = 0 \implies x = 2$$

A perfect square — there is exactly one repeated solution, meaning the line is tangent to the rectangular hyperbola.

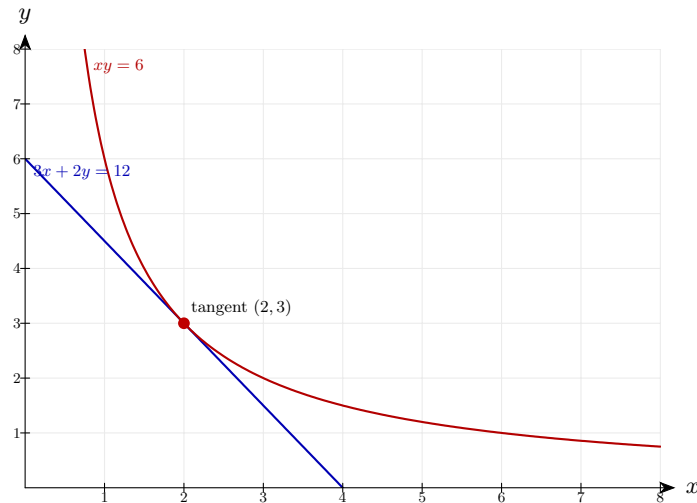
Step 5: Find y .

$$y = \frac{12 - 6}{2} = 3$$

Substitute $x = 2$ into the rearranged linear.

Final answer: $(x, y) = (2, 3)$; the line is tangent to the hyperbola $xy = 6$ at this point.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 35 (ise5-035)

Difficulty: Standard / Marks: 6

Find the points of intersection of the line $y = 2x - 1$ with the circle $x^2 + y^2 = 25$.

Worked solution.

Step 1: Substitute $y = 2x - 1$ into the circle.

$$x^2 + (2x - 1)^2 = 25$$

Bracket the linear expression before squaring to avoid sign errors.

Step 2: Expand.

$$x^2 + 4x^2 - 4x + 1 = 25 \implies 5x^2 - 4x - 24 = 0$$

$$(2x - 1)^2 = 4x^2 - 4x + 1.$$

Step 3: Solve the quadratic.

$$x = \frac{4 \pm \sqrt{16 + 480}}{10} = \frac{4 \pm \sqrt{496}}{10}$$

Discriminant $16 + 480 = 496 = 16 \times 31$, so $\sqrt{496} = 4\sqrt{31}$.

Step 4: Simplify.

$$x = \frac{4 \pm 4\sqrt{31}}{10} = \frac{2 \pm 2\sqrt{31}}{5}$$

Divide numerator and denominator by 2.

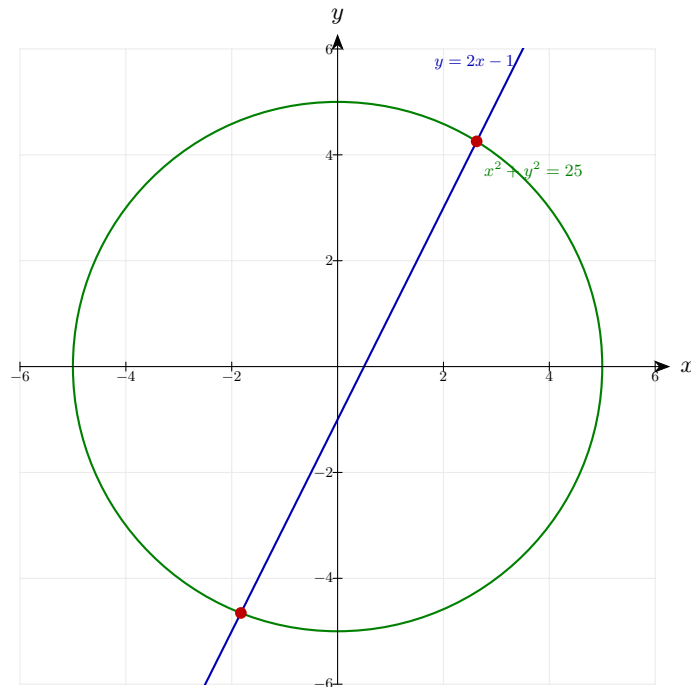
Step 5: Find each y from $y = 2x - 1$.

$$y = 2 \cdot \frac{2 \pm 2\sqrt{31}}{5} - 1 = \frac{4 \pm 4\sqrt{31}}{5} - \frac{5}{5} = \frac{-1 \pm 4\sqrt{31}}{5}$$

Combine over a common denominator.

Final answer: $(x, y) = \left(\frac{2 + 2\sqrt{31}}{5}, \frac{-1 + 4\sqrt{31}}{5} \right)$ and $\left(\frac{2 - 2\sqrt{31}}{5}, \frac{-1 - 4\sqrt{31}}{5} \right)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 36 (ise5-036)

Difficulty: Standard / Marks: 6

Find the points where the line $x + 2y = 5$ meets the circle $x^2 + y^2 = 10$.

Worked solution.

Step 1: Rearrange the linear equation.

$$x = 5 - 2y$$

Make x the subject.

Step 2: Substitute into the circle.

$$(5 - 2y)^2 + y^2 = 10$$

Replace x with $5 - 2y$ in the circle equation.

Step 3: Expand and simplify.

$$25 - 20y + 4y^2 + y^2 = 10 \implies 5y^2 - 20y + 15 = 0 \implies y^2 - 4y + 3 = 0$$

Divide through by 5 for friendlier numbers.

Step 4: Factorise.

$$(y - 1)(y - 3) = 0 \implies y = 1 \text{ or } y = 3$$

Product 3, sum -4 : $-1, -3$.

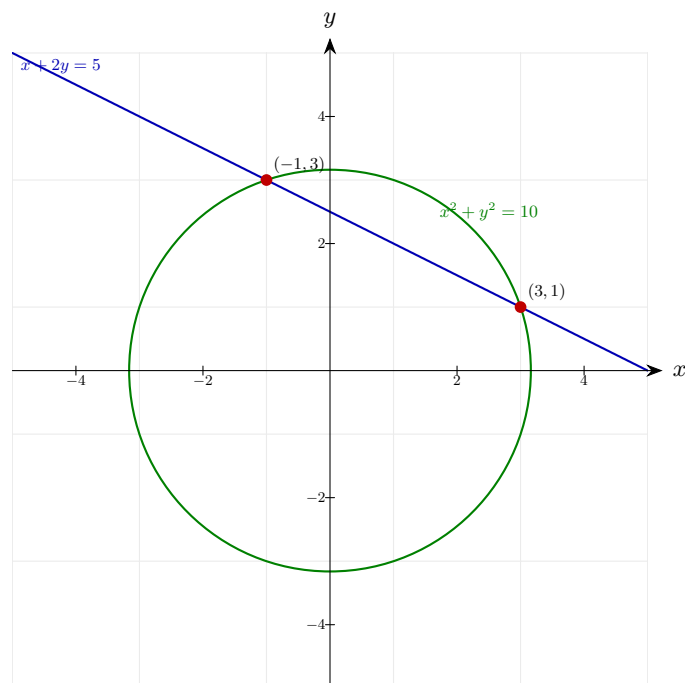
Step 5: Find each x from $x = 5 - 2y$.

$$y = 1 \Rightarrow x = 3; \quad y = 3 \Rightarrow x = -1$$

Sanity check: $3^2 + 1^2 = 10\checkmark$, $(-1)^2 + 3^2 = 10\checkmark$.

Final answer: $(x, y) = (3, 1)$ and $(x, y) = (-1, 3)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 37 (ise5-037)

Difficulty: Standard / Marks: 6

Find the intersection points of the line $y = x$ with the circle $(x - 2)^2 + (y - 1)^2 = 5$.

Worked solution.

Step 1: Substitute $y = x$ into the circle.

$$(x - 2)^2 + (x - 1)^2 = 5$$

Both x and y appear in the brackets — replacing y with x is straightforward.

Step 2: Expand.

$$x^2 - 4x + 4 + x^2 - 2x + 1 = 5 \implies 2x^2 - 6x = 0$$

Combine like terms and move the constant to the left.

Step 3: Factor and solve.

$$2x(x - 3) = 0 \implies x = 0 \text{ or } x = 3$$

Factor out $2x$ from both terms.

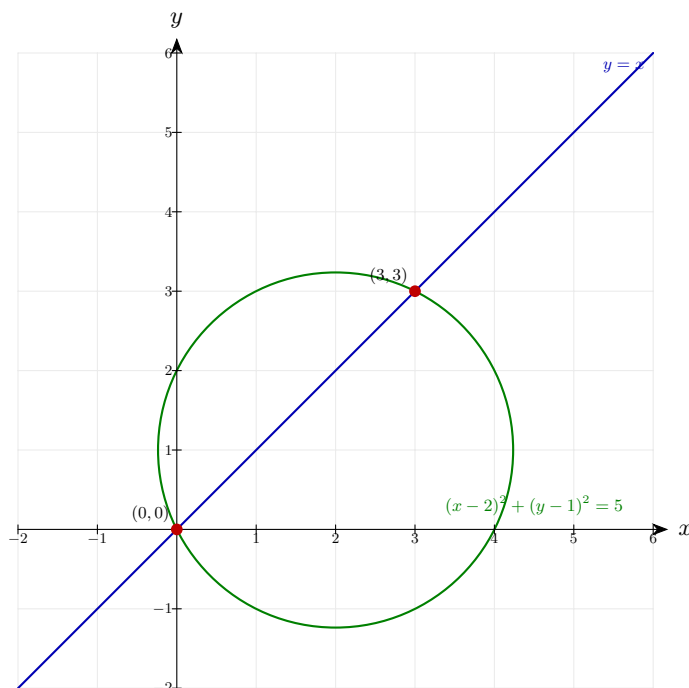
Step 4: Find each y using $y = x$.

$$x = 0 \Rightarrow y = 0; \quad x = 3 \Rightarrow y = 3$$

Verify: $(0 - 2)^2 + (0 - 1)^2 = 4 + 1 = 5\checkmark$ and $(3 - 2)^2 + (3 - 1)^2 = 1 + 4 = 5\checkmark$.

Final answer: $(x, y) = (0, 0)$ and $(x, y) = (3, 3)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 38 (ise5-038)

Difficulty: Standard / Marks: 6

Find the points of intersection of the line $y = -x + 4$ with the circle $x^2 + y^2 = 8$.

Worked solution.

Step 1: Substitute the line into the circle.

$$x^2 + (-x + 4)^2 = 8$$

Replace y with $-x + 4$ and square the whole thing.

Step 2: Expand.

$$x^2 + x^2 - 8x + 16 = 8 \implies 2x^2 - 8x + 8 = 0 \implies x^2 - 4x + 4 = 0$$

$(-x + 4)^2 = x^2 - 8x + 16$. Divide by 2 to simplify.

Step 3: Solve.

$$(x - 2)^2 = 0 \implies x = 2$$

A repeated root — the line is tangent to the circle.

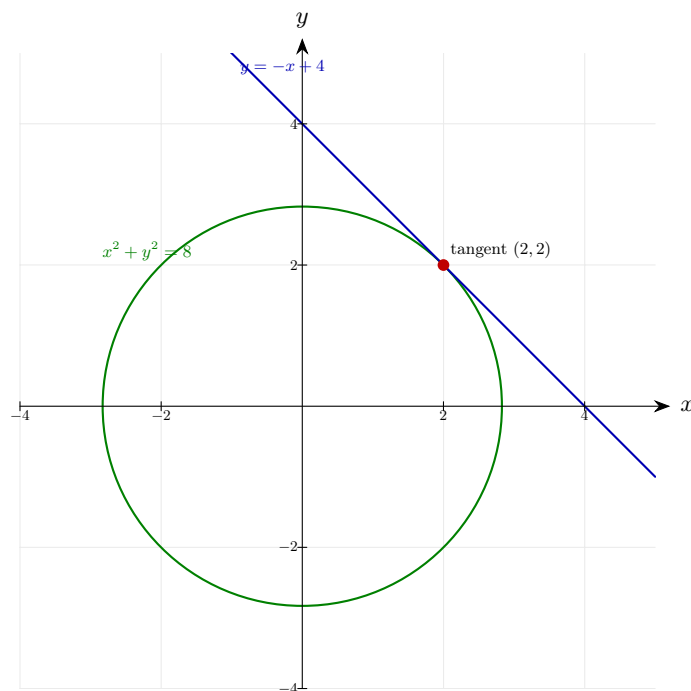
Step 4: Find y .

$$y = -2 + 4 = 2$$

Substitute into the linear equation.

Final answer: The line is tangent to the circle at $(2, 2)$.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 39 (ise5-039)

Difficulty: Challenge / Marks: 7

Find the value of k such that the line $y = 2x + k$ is tangent to the curve $y = x^2$. State the point of contact.

Worked solution.

Step 1: Set up the simultaneous equations and equate.

$$x^2 = 2x + k \implies x^2 - 2x - k = 0$$

For a tangent, the resulting quadratic in x must have a single (repeated) root — i.e. discriminant zero.

Step 2: Set the discriminant to zero.

$$b^2 - 4ac = 4 + 4k = 0 \implies k = -1$$

Tangency is the algebraic statement "discriminant = 0".

Step 3: Find the tangent point by solving the (now perfect-square) quadratic.

$$x^2 - 2x + 1 = 0 \implies (x - 1)^2 = 0 \implies x = 1$$

Substitute $k = -1$ back into the quadratic. The repeated root is the x -coordinate of the tangent point.

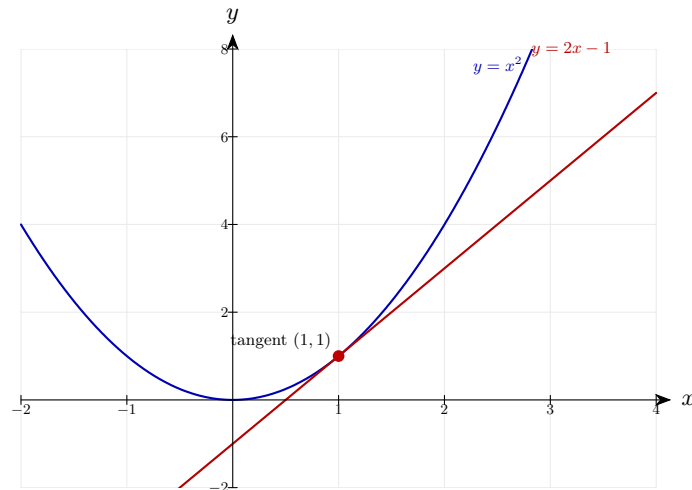
Step 4: Find y .

$$y = 2(1) + (-1) = 1$$

Substitute $x = 1$ into the linear (with $k = -1$).

Final answer: $k = -1$; the tangent point is $(1, 1)$.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 40 (ise5-040)

Difficulty: Challenge / Marks: 7

Find the values of k for which the line $y = x + k$ is tangent to the circle $x^2 + y^2 = 8$. State each point of contact.

Worked solution.

Step 1: Substitute the line into the circle.

$$x^2 + (x + k)^2 = 8$$

Replace y with $x + k$ in the circle.

Step 2: Expand and simplify.

$$x^2 + x^2 + 2kx + k^2 = 8 \implies 2x^2 + 2kx + (k^2 - 8) = 0$$

A quadratic in x with coefficients depending on k .

Step 3: Apply the tangency condition: discriminant zero.

$$(2k)^2 - 4(2)(k^2 - 8) = 0 \implies 4k^2 - 8k^2 + 64 = 0 \implies -4k^2 = -64 \implies k = \pm 4$$

Two tangent lines exist — one above the circle and one below — corresponding to the two values of k .

Step 4: Find the tangent points. For $k = 4$:

$$2x^2 + 8x + 8 = 0 \implies x^2 + 4x + 4 = 0 \implies (x + 2)^2 = 0 \implies x = -2, y = 2$$

A perfect-square trinomial with repeated root $x = -2$, giving the tangent point $(-2, 2)$.

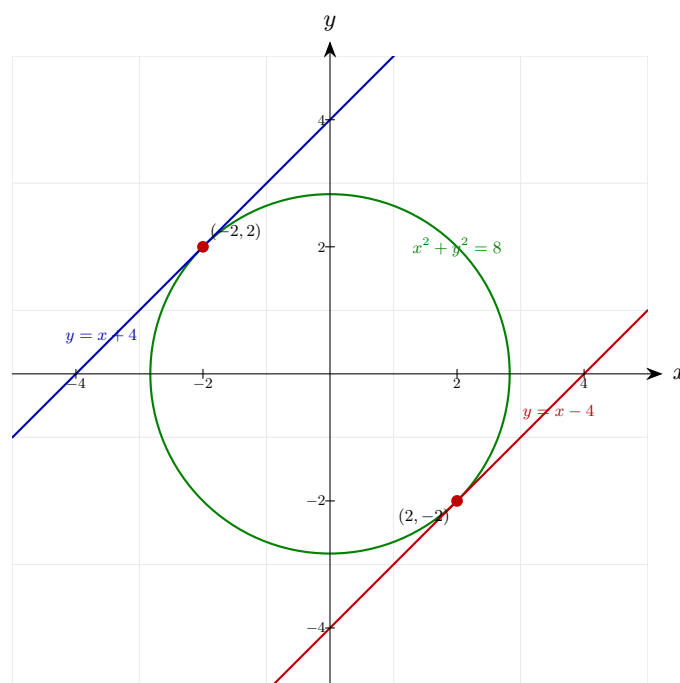
Step 5: For $k = -4$:

$$2x^2 - 8x + 8 = 0 \implies (x - 2)^2 = 0 \implies x = 2, y = -2$$

Tangent point $(2, -2)$. The two tangent points are diametrically opposite on the circle.

Final answer: $k = 4$; tangent at $(-2, 2)$. $k = -4$; tangent at $(2, -2)$.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 41 (ise5-041)

Difficulty: Challenge / Marks: 7

The line $y = mx + 1$ is tangent to the curve $y = x^2 + 2$. Find the possible values of m and the corresponding points of contact.

Worked solution.

Step 1: Equate the two expressions.

$$mx + 1 = x^2 + 2 \implies x^2 - mx + 1 = 0$$

Set the right-hand sides equal and rearrange into the standard quadratic.

Step 2: Apply the tangency condition.

$$(-m)^2 - 4(1)(1) = 0 \implies m^2 = 4 \implies m = \pm 2$$

Discriminant zero gives two valid gradients — there are exactly two lines through $(0, 1)$ tangent to the parabola.

Step 3: For $m = 2$, find the tangent point.

$$x^2 - 2x + 1 = 0 \implies (x - 1)^2 = 0 \implies x = 1, y = 2(1) + 1 = 3$$

Substitute and solve the perfect-square equation; tangent point $(1, 3)$.

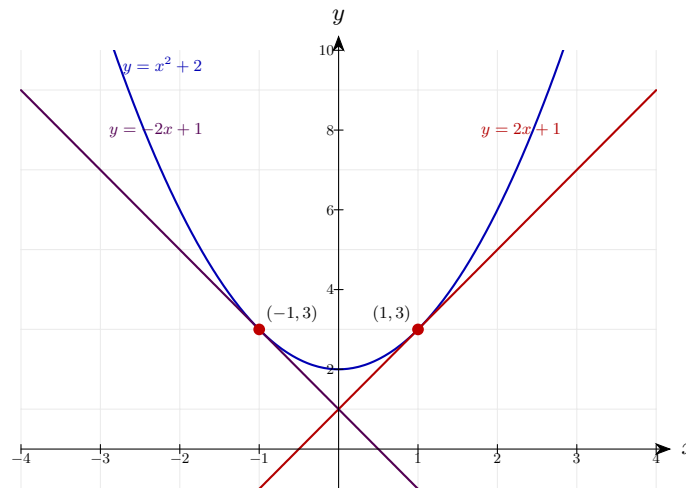
Step 4: For $m = -2$:

$$x^2 + 2x + 1 = 0 \implies (x + 1)^2 = 0 \implies x = -1, y = -2(-1) + 1 = 3$$

Tangent point $(-1, 3)$. By symmetry of $y = x^2 + 2$ about the y -axis, the two tangent points have the same y -coordinate.

Final answer: $m = 2$; tangent at $(1, 3)$. $m = -2$; tangent at $(-1, 3)$.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 42 (ise5-042)

Difficulty: Challenge / Marks: 6

Find the range of values of k for which the line $y = 2x + k$ meets the curve $y = x^2 + 1$ at two distinct points.

Worked solution.

Step 1: Equate and rearrange.

$$2x + k = x^2 + 1 \implies x^2 - 2x + (1 - k) = 0$$

A quadratic in x whose coefficients depend on k .

Step 2: For two distinct intersection points, demand discriminant > 0 .

$$(-2)^2 - 4(1)(1 - k) > 0 \implies 4 - 4 + 4k > 0 \implies 4k > 0 \implies k > 0$$

Two distinct real roots correspond to two distinct intersection points. Discriminant > 0 is exactly this condition.

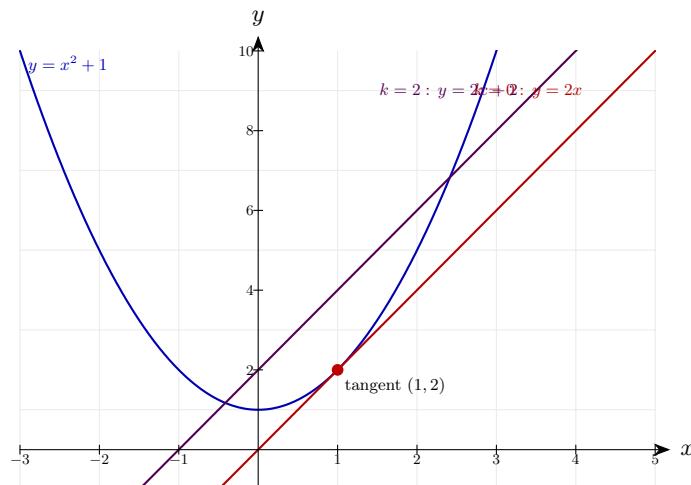
Step 3: Interpret.

$$k > 0$$

When $k = 0$ the line is tangent (one intersection); when $k < 0$ it misses the curve; when $k > 0$ it cuts twice.

Final answer: $k > 0$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 43 (ise5-043)

Difficulty: Challenge / Marks: 5

Find the points of intersection of the curves $y = x^2$ and $y = 4 - x^2$.

Worked solution.

Step 1: Set the two expressions for y equal.

$$x^2 = 4 - x^2 \implies 2x^2 = 4 \implies x^2 = 2$$

Even though both equations are quadratic, the standard technique still works: equate the right-hand sides.

Step 2: Solve for x .

$$x = \pm\sqrt{2}$$

Take square roots, remembering both the positive and negative.

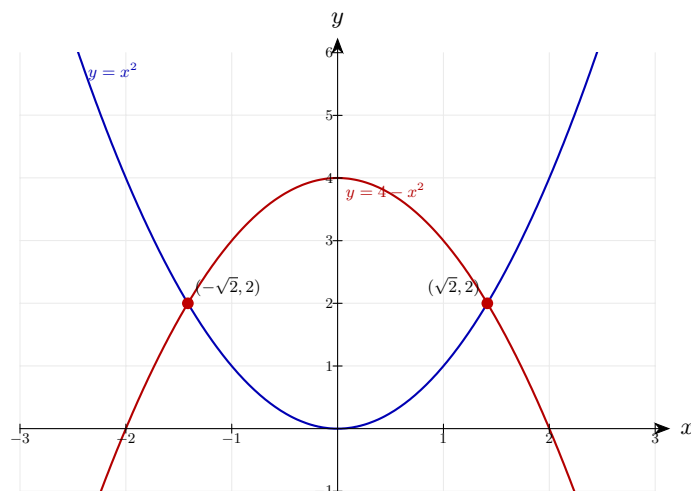
Step 3: Find each y .

$$y = x^2 = 2$$

Either curve will give the same value of y for each x — that is the whole point of an intersection.

Final answer: $(x, y) = (\sqrt{2}, 2)$ and $(x, y) = (-\sqrt{2}, 2)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 44 (ise5-044)

Difficulty: Challenge / Marks: 6

Find the points of intersection of the parabolas $y = x^2 - 2x$ and $y = -x^2 + 4x - 2$.

Worked solution.

Step 1: Equate the right-hand sides.

$$x^2 - 2x = -x^2 + 4x - 2$$

Set the two expressions for y equal.

Step 2: Rearrange.

$$2x^2 - 6x + 2 = 0 \implies x^2 - 3x + 1 = 0$$

Add x^2 and subtract $4x - 2$; then divide through by 2.

Step 3: Apply the quadratic formula.

$$x = \frac{3 \pm \sqrt{9 - 4}}{2} = \frac{3 \pm \sqrt{5}}{2}$$

Discriminant 5 is not a perfect square, so the roots are irrational.

Step 4: Find each y — use the simpler curve $y = x^2 - 2x = x(x - 2)$.

$$x = \frac{3 + \sqrt{5}}{2} \Rightarrow x - 2 = \frac{-1 + \sqrt{5}}{2}, y = \frac{(3 + \sqrt{5})(-1 + \sqrt{5})}{4} = \frac{-3 + 3\sqrt{5} - \sqrt{5} + 5}{4} = \frac{2 + 2\sqrt{5}}{4} = \frac{1 + \sqrt{5}}{2}$$

Expand the product carefully, then simplify.

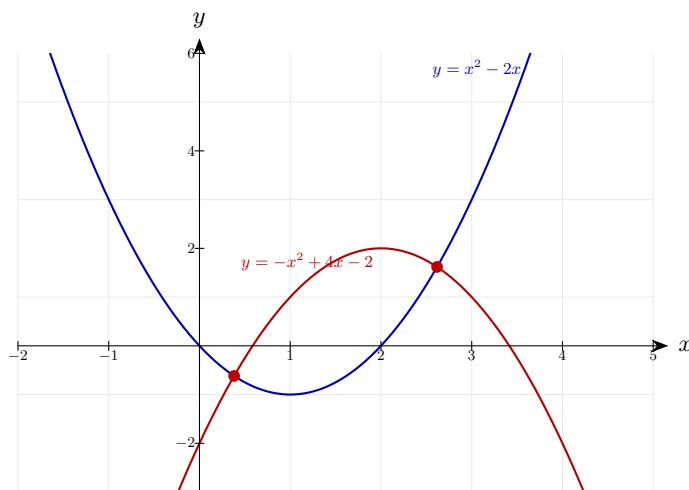
Step 5: Similarly for the other root.

$$x = \frac{3 - \sqrt{5}}{2} \Rightarrow y = \frac{1 - \sqrt{5}}{2}$$

Replace $\sqrt{5}$ by $-\sqrt{5}$ throughout the previous computation.

Final answer: $(x, y) = \left(\frac{3 + \sqrt{5}}{2}, \frac{1 + \sqrt{5}}{2}\right)$ and $\left(\frac{3 - \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2}\right)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 45 (ise5-045)

Difficulty: Challenge / Marks: 7

Find the points where the parabola $y = x^2 - 3$ meets the circle $x^2 + y^2 = 9$.

Worked solution.

Step 1: Substitute $y = x^2 - 3$ into the circle.

$$x^2 + (x^2 - 3)^2 = 9$$

Replace y in the circle equation. This is genuinely a quartic in x .

Step 2: Expand.

$$x^2 + x^4 - 6x^2 + 9 = 9 \implies x^4 - 5x^2 = 0$$

Expand $(x^2 - 3)^2 = x^4 - 6x^2 + 9$ and cancel the constants.

Step 3: Factor.

$$x^2(x^2 - 5) = 0 \implies x = 0 \text{ or } x = \pm\sqrt{5}$$

Treat as a quadratic in x^2 — or factor out x^2 directly. Three distinct x -values (with $x = 0$ repeated).

Step 4: Find each y using $y = x^2 - 3$.

$$x = 0 \Rightarrow y = -3; \quad x = \pm\sqrt{5} \Rightarrow y = 5 - 3 = 2$$

Now we have three intersection points: $(0, -3)$, $(\sqrt{5}, 2)$, $(-\sqrt{5}, 2)$.

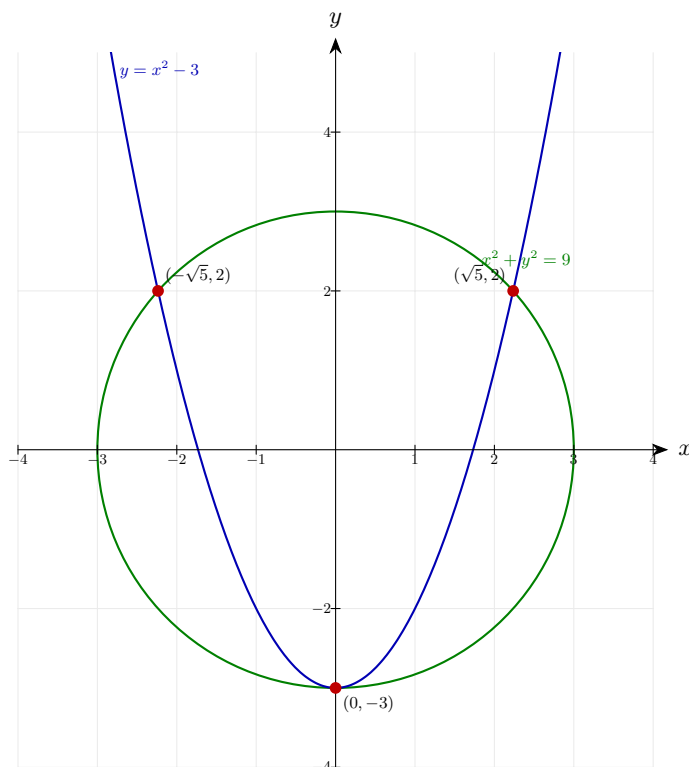
Step 5: Verify on the circle.

$$0^2 + (-3)^2 = 9\checkmark; \quad 5 + 4 = 9\checkmark$$

All three points lie on the circle. The parabola is tangent to the circle at the bottom point and crosses it twice further up.

Final answer: $(x, y) = (0, -3), (\sqrt{5}, 2)$ and $(-\sqrt{5}, 2)$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 46 (ise5-046)

Difficulty: Challenge / Marks: 6

A rectangle has perimeter 22 cm and area 28 cm². Find the lengths of its sides.

Worked solution.

Step 1: Let x and y denote the side lengths and write down the two given conditions.

$$2(x + y) = 22 \implies x + y = 11; \quad xy = 28$$

A rectangle of sides x and y has perimeter $2(x + y)$ and area xy .

Step 2: Make y the subject of the linear equation.

$$y = 11 - x$$

This lets you substitute into the area equation.

Step 3: Substitute and form a quadratic.

$$x(11 - x) = 28 \implies 11x - x^2 = 28 \implies x^2 - 11x + 28 = 0$$

Move everything to one side so the leading coefficient is positive.

Step 4: Factorise.

$$(x - 4)(x - 7) = 0 \implies x = 4 \text{ or } x = 7$$

Product 28, sum -11 : $-4, -7$.

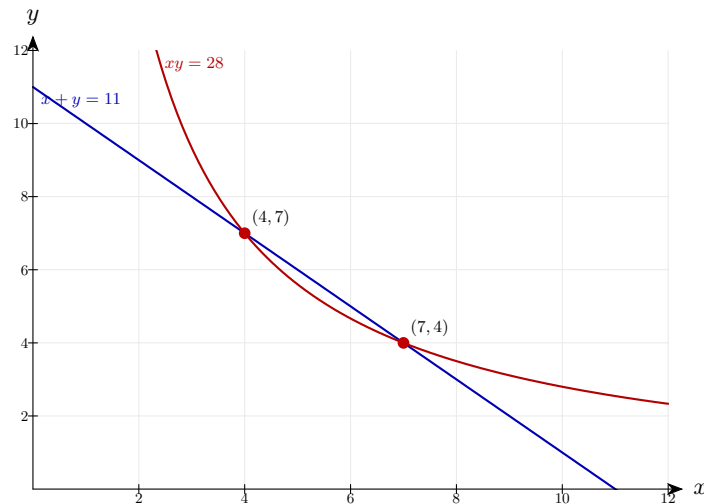
Step 5: Find the partner length.

$$x = 4 \Rightarrow y = 7; \quad x = 7 \Rightarrow y = 4$$

The two solutions correspond to the same rectangle with the role of "length" and "width" swapped.

Final answer: The sides are 4 cm and 7 cm.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 47 (ise5-047)

Difficulty: Challenge / Marks: 6

The sum of two positive numbers is 12 and the sum of their squares is 80. Find the two numbers.

Worked solution.

Step 1: Let the numbers be x and y , and translate the conditions.

$$x + y = 12; \quad x^2 + y^2 = 80$$

One linear, one quadratic — classic structure for substitution.

Step 2: Rearrange the linear equation.

$$y = 12 - x$$

Make y the subject so it can be substituted into the quadratic.

Step 3: Substitute.

$$x^2 + (12 - x)^2 = 80$$

Bracket the linear expression before squaring.

Step 4: Expand and simplify.

$$x^2 + 144 - 24x + x^2 = 80 \implies 2x^2 - 24x + 64 = 0 \implies x^2 - 12x + 32 = 0$$

Divide through by 2 to simplify.

Step 5: Factorise.

$$(x - 4)(x - 8) = 0 \implies x = 4 \text{ or } x = 8$$

Product 32, sum -12 : -4 and -8 .

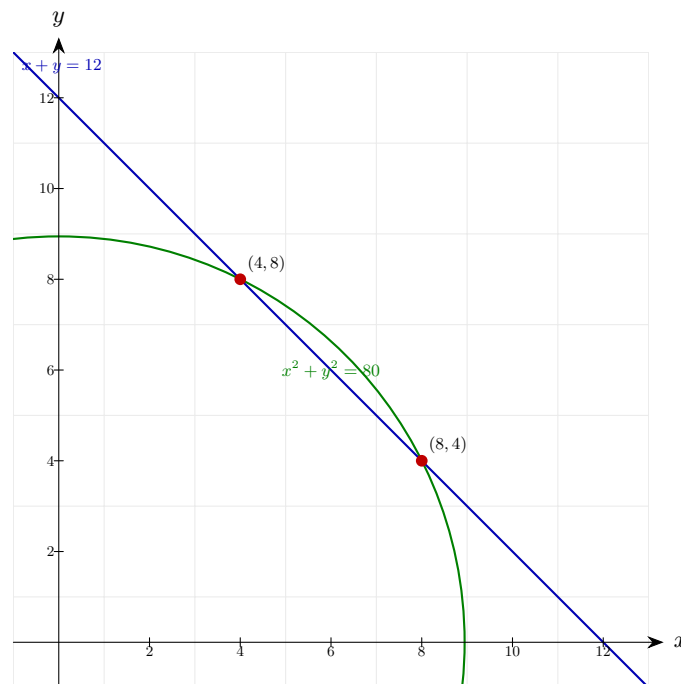
Step 6: Find each partner number.

$$x = 4 \Rightarrow y = 8; \quad x = 8 \Rightarrow y = 4$$

Both solutions describe the same unordered pair of numbers.

Final answer: The two numbers are 4 and 8.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 48 (ise5-048)

Difficulty: Challenge / Marks: 7

A right-angled triangle has hypotenuse 13 cm and perimeter 30 cm. Find the lengths of the two shorter sides.

Worked solution.

Step 1: Let the two shorter sides be x and y . Translate the conditions.

$$x + y + 13 = 30 \implies x + y = 17; \quad x^2 + y^2 = 13^2 = 169$$

The perimeter equation is linear and Pythagoras gives the quadratic constraint.

Step 2: Make y the subject of the linear equation.

$$y = 17 - x$$

Ready for substitution.

Step 3: Substitute into Pythagoras.

$$x^2 + (17 - x)^2 = 169$$

Bracket before squaring.

Step 4: Expand and simplify.

$$x^2 + 289 - 34x + x^2 = 169 \implies 2x^2 - 34x + 120 = 0 \implies x^2 - 17x + 60 = 0$$

Divide through by 2.

Step 5: Factorise.

$$(x - 5)(x - 12) = 0 \implies x = 5 \text{ or } x = 12$$

Product 60, sum -17 : -5 and -12 .

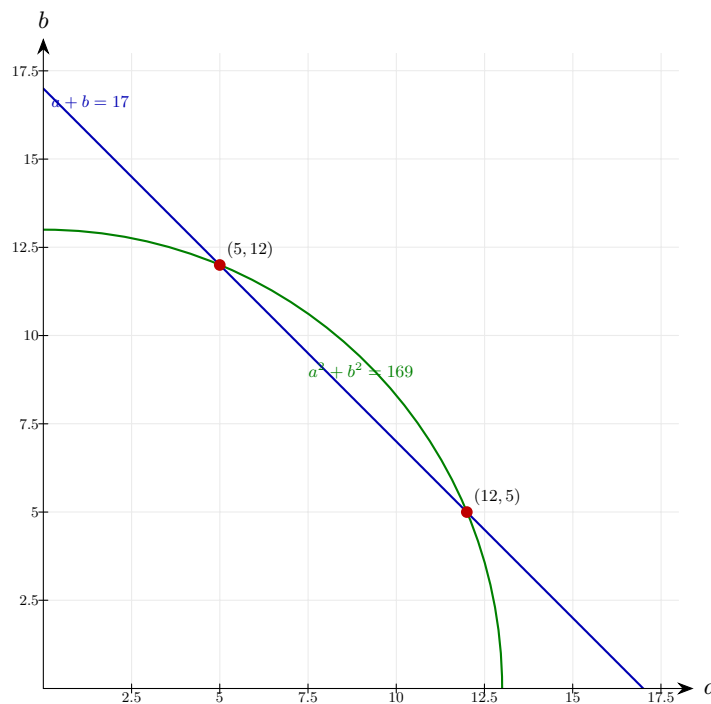
Step 6: Find the partner side and check.

$$x = 5 \Rightarrow y = 12; \quad 5^2 + 12^2 = 25 + 144 = 169 \checkmark$$

This is the classic 5-12-13 Pythagorean triple. Both solutions describe the same triangle, just with the labels of the legs swapped.

Final answer: The two shorter sides are 5 cm and 12 cm.

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 49 (ise5-049)

Difficulty: Challenge / *Marks: 7*

The line $y = x + 2$ meets the curve $y = x^2 - 4x + 8$ at points A and B . Find the coordinates of A and B , and hence the length AB in exact form.

Worked solution.

Step 1: Equate the two expressions for y .

$$x + 2 = x^2 - 4x + 8$$

Set the right-hand sides equal at the intersection points.

Step 2: Rearrange.

$$x^2 - 5x + 6 = 0$$

$$-4x - x = -5x, 8 - 2 = 6.$$

Step 3: Factorise.

$$(x - 2)(x - 3) = 0 \implies x = 2 \text{ or } x = 3$$

Product 6, sum -5 .

Step 4: Find each y and so the coordinates.

$$x = 2 \Rightarrow y = 4; \quad x = 3 \Rightarrow y = 5$$

So $A = (2, 4)$ and $B = (3, 5)$ (the labelling is arbitrary).

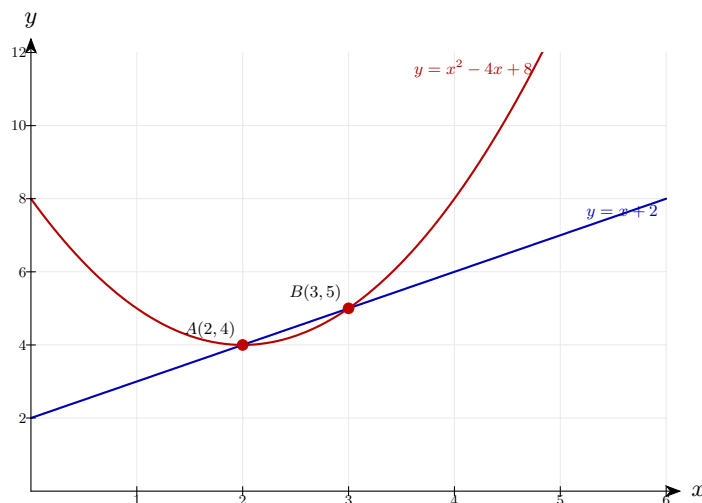
Step 5: Compute AB using the distance formula.

$$AB = \sqrt{(3 - 2)^2 + (5 - 4)^2} = \sqrt{1 + 1} = \sqrt{2}$$

$AB^2 = (\Delta x)^2 + (\Delta y)^2$. Leave the answer in exact surd form unless asked otherwise.

Final answer: $A = (2, 4)$, $B = (3, 5)$; $AB = \sqrt{2}$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.

Simultaneous Equations (Quadratic) 50 (ise5-050)

Difficulty: Challenge / Marks: 8

The line $y = 2x + 1$ meets the circle $x^2 + y^2 = 10$ at points P and Q . Find the coordinates of P and Q , the midpoint M of PQ , and the length PQ in exact form.

Worked solution.

Step 1: Substitute the line into the circle.

$$x^2 + (2x + 1)^2 = 10$$

Replace y with $2x + 1$.

Step 2: Expand and simplify.

$$x^2 + 4x^2 + 4x + 1 = 10 \implies 5x^2 + 4x - 9 = 0$$

$(2x + 1)^2 = 4x^2 + 4x + 1$, then move everything to one side.

Step 3: Factorise.

$$(5x + 9)(x - 1) = 0 \implies x = -\frac{9}{5} \text{ or } x = 1$$

Check the expansion: $5x \cdot x = 5x^2$, $5x \cdot (-1) + 9 \cdot x = -5x + 9x = 4x$, $9 \cdot (-1) = -9$. Good.

Step 4: Find each y from $y = 2x + 1$.

$$x = 1 \Rightarrow y = 3; \quad x = -\frac{9}{5} \Rightarrow y = -\frac{13}{5}$$

$$2 \cdot \left(-\frac{9}{5}\right) + 1 = -\frac{18}{5} + \frac{5}{5} = -\frac{13}{5}.$$

Step 5: Compute the midpoint M of PQ .

$$M = \left(\frac{1 + \left(-\frac{9}{5}\right)}{2}, \frac{3 + \left(-\frac{13}{5}\right)}{2} \right) = \left(\frac{-\frac{4}{5}}{2}, \frac{\frac{2}{5}}{2} \right) = \left(-\frac{2}{5}, \frac{1}{5} \right)$$

Midpoint is the average of the coordinates. Combine over common denominators first.

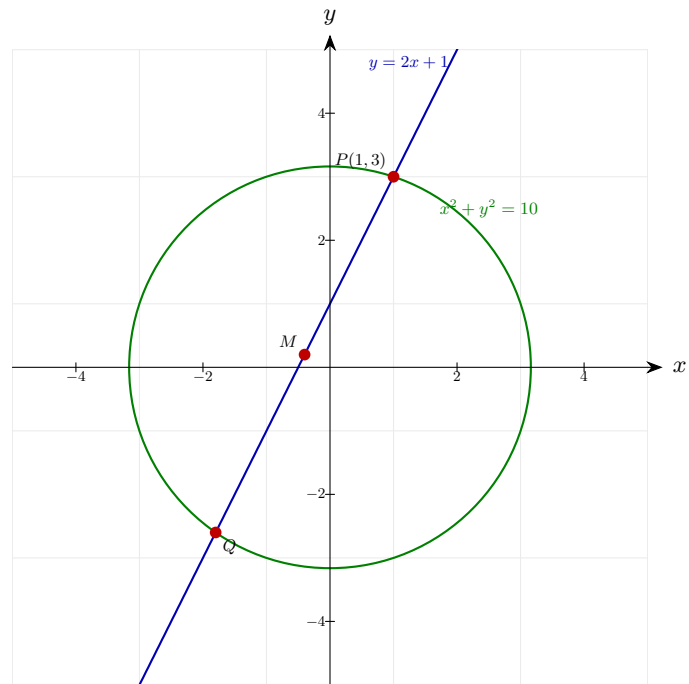
Step 6: Compute PQ using the distance formula.

$$PQ = \sqrt{\left(1 + \frac{9}{5}\right)^2 + \left(3 + \frac{13}{5}\right)^2} = \sqrt{\left(\frac{14}{5}\right)^2 + \left(\frac{28}{5}\right)^2} = \sqrt{\frac{196 + 784}{25}} = \sqrt{\frac{980}{25}} = \frac{\sqrt{980}}{5} = \frac{14\sqrt{5}}{5}$$

Note $\frac{28}{5} = 2 \cdot \frac{14}{5}$, so the second component is twice the first — consistent with a line of gradient 2. Simplify $\sqrt{980} = \sqrt{196 \cdot 5} = 14\sqrt{5}$.

Final answer: $P = (1, 3)$, $Q = \left(-\frac{9}{5}, -\frac{13}{5}\right)$; midpoint $M = \left(-\frac{2}{5}, \frac{1}{5}\right)$; $PQ = \frac{14\sqrt{5}}{5}$

Diagram.



The two equations plotted, with the solution shown at the point(s) of intersection.
